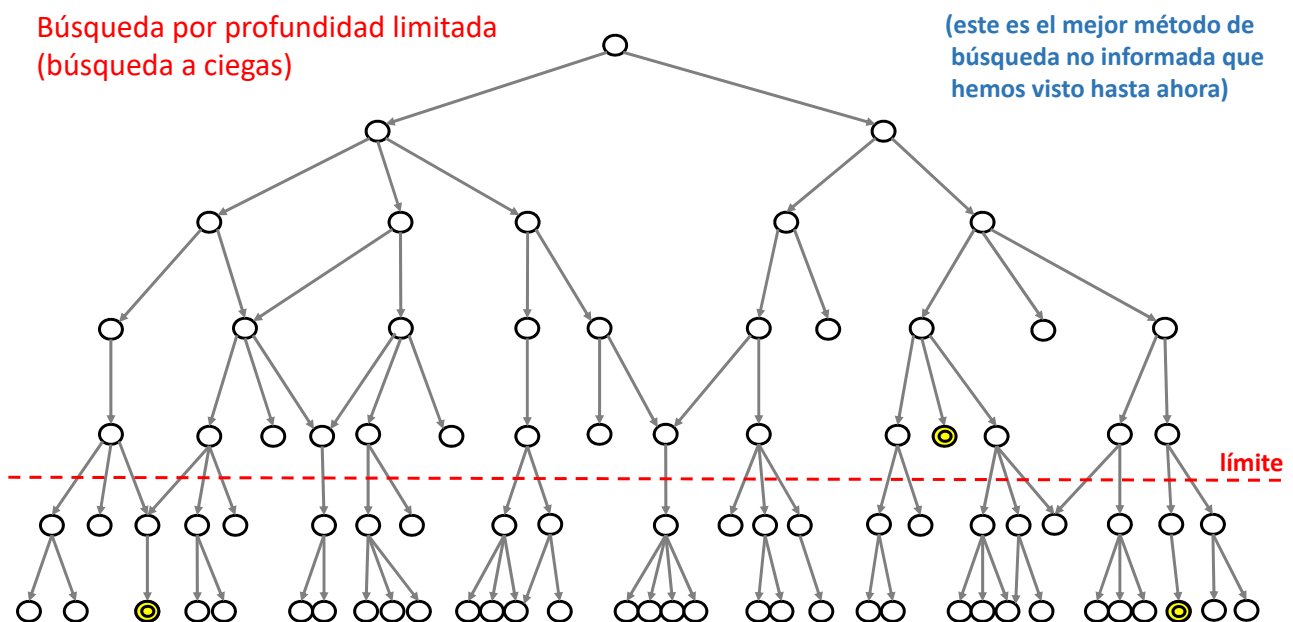


# Búsqueda informada: Ejemplos de *Branch & Bound* y *Best First Search*

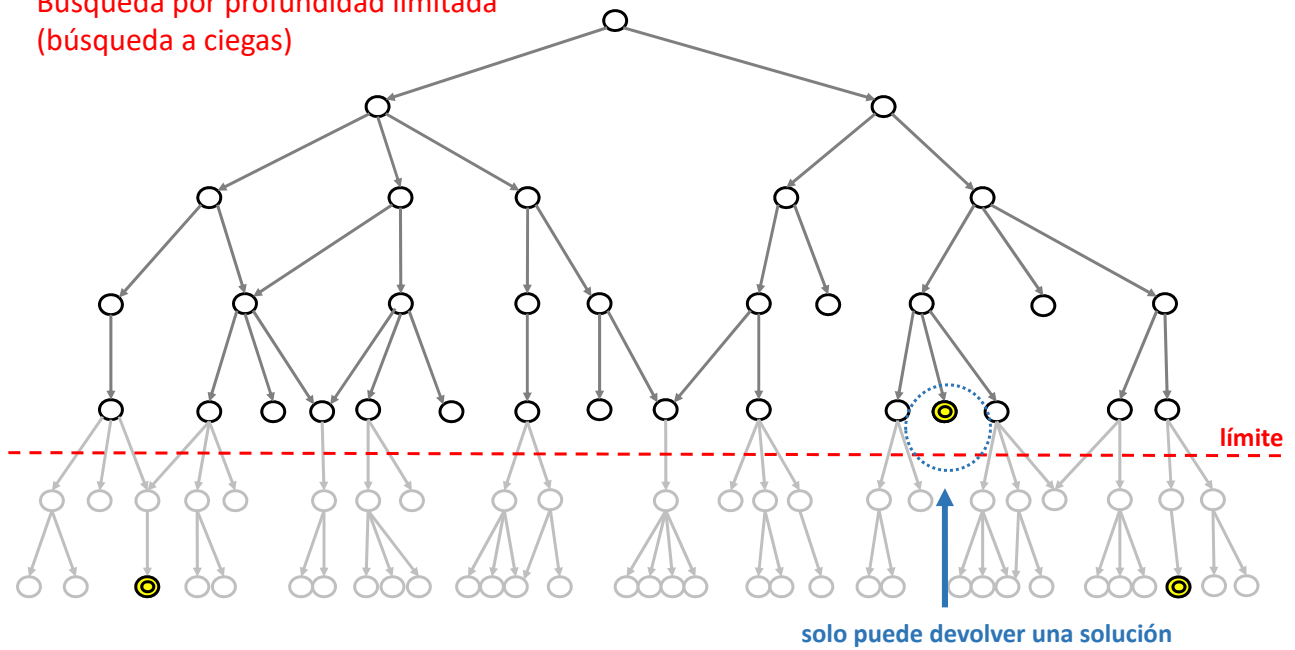
Javier Vázquez

CS – FIB- UPC

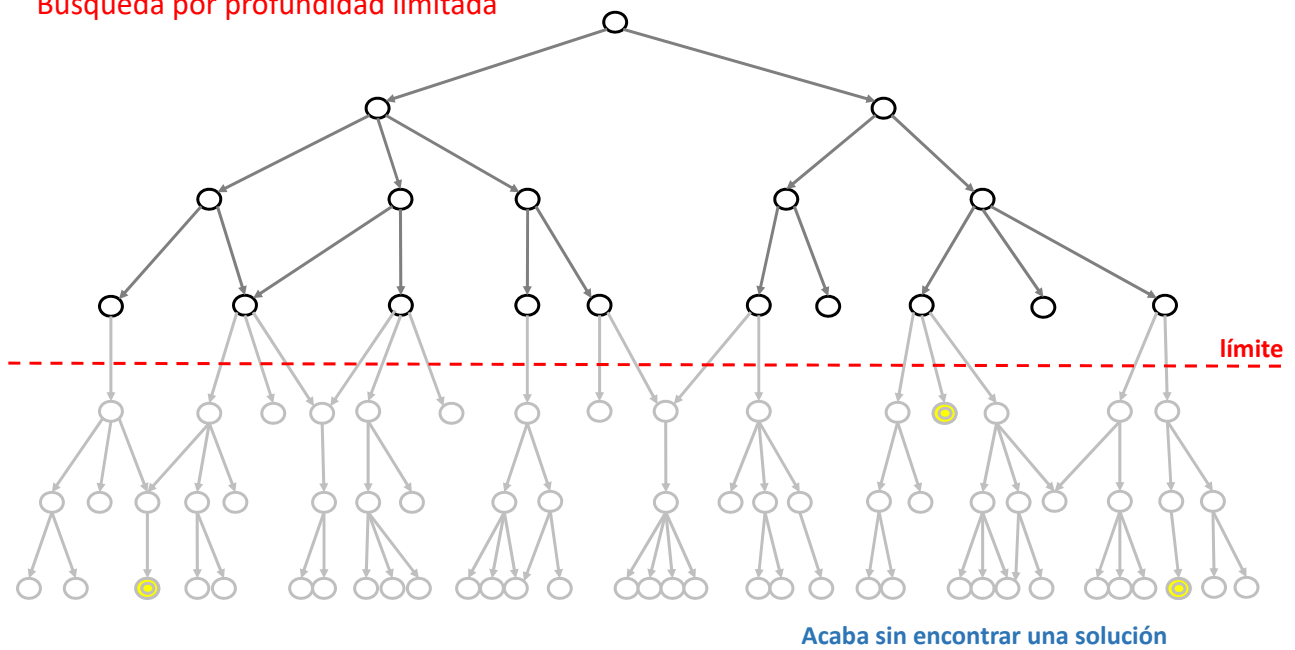
1



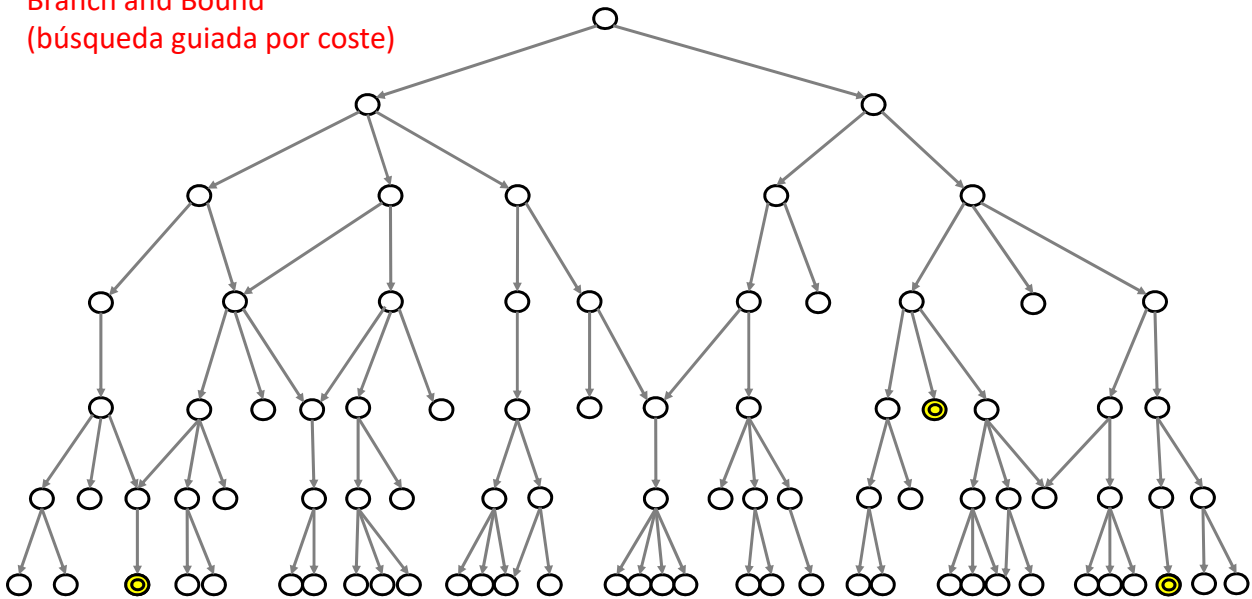
Búsqueda por profundidad limitada  
(búsqueda a ciegas)



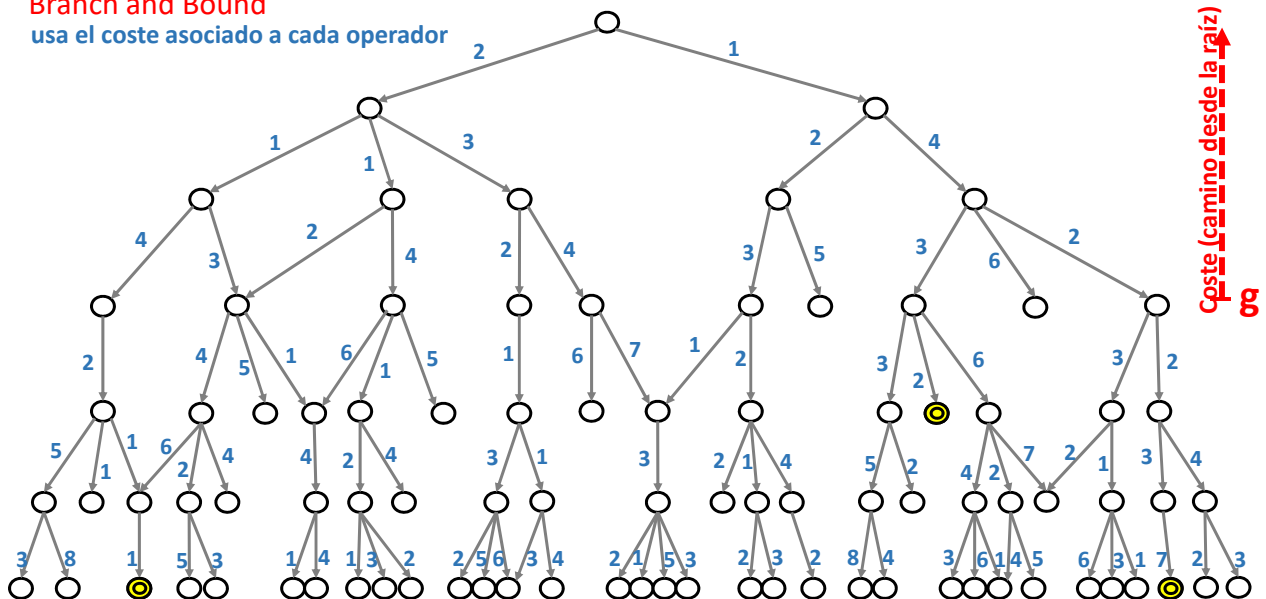
Búsqueda por profundidad limitada



Branch and Bound  
(búsqueda guiada por coste)



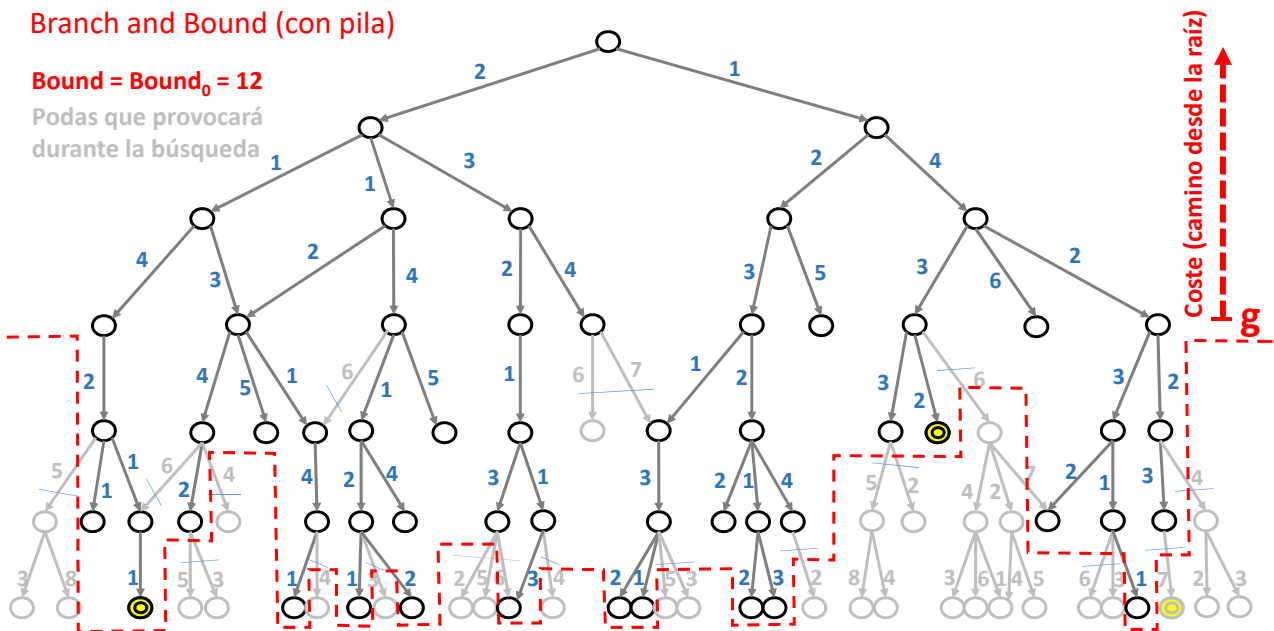
Branch and Bound  
usa el coste asociado a cada operador



## Branch and Bound (con pila)

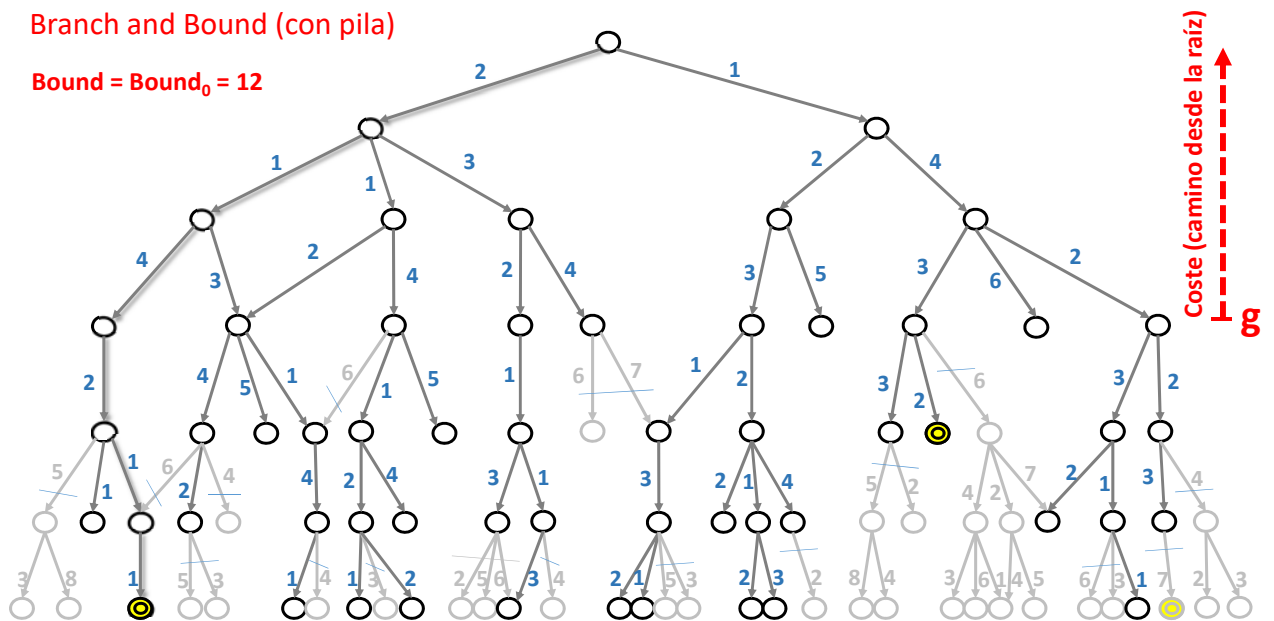
**Bound = Bound<sub>0</sub> = 12**

Podas que provocará durante la búsqueda



No hay un límite fijo de nivel, paramos en cada rama a diferentes niveles según el límite de coste (bound)

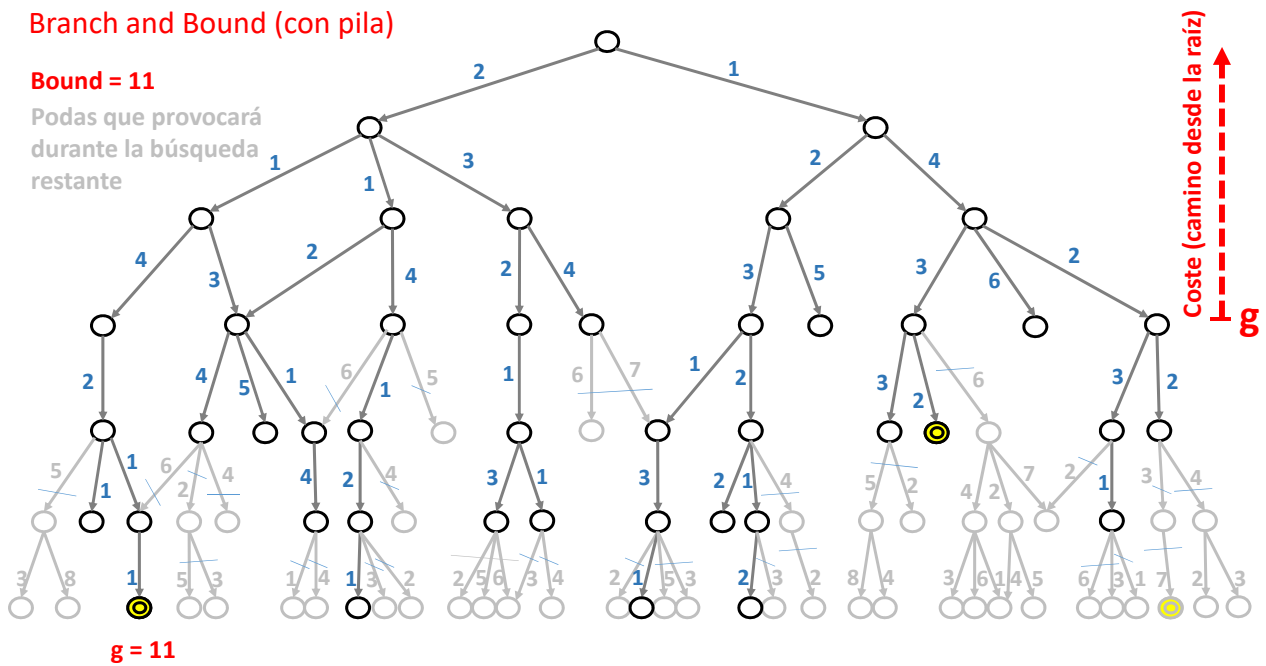
## Branch and Bound (con pila)

**Bound = Bound<sub>0</sub> = 12****g = 11** ← El coste de la solución es menor que el bound → **Bound = 11**

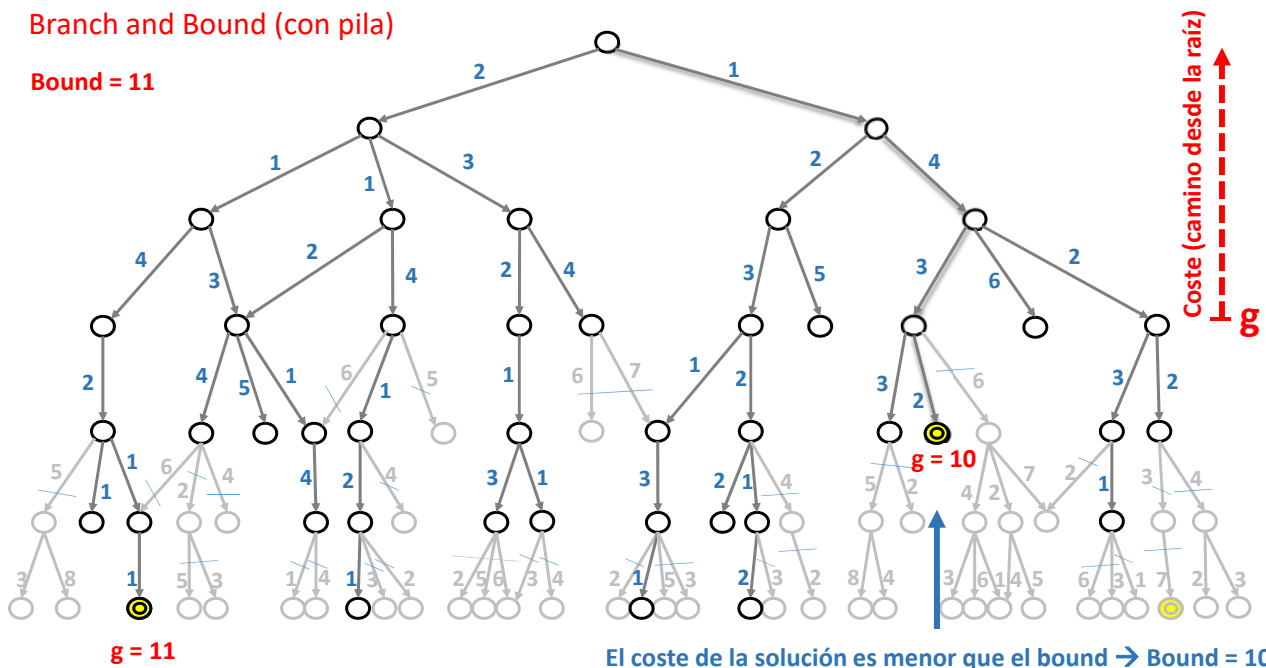
## Branch and Bound (con pila)

**Bound = 11**

Podas que provocará durante la búsqueda restante



## Branch and Bound (con pila)

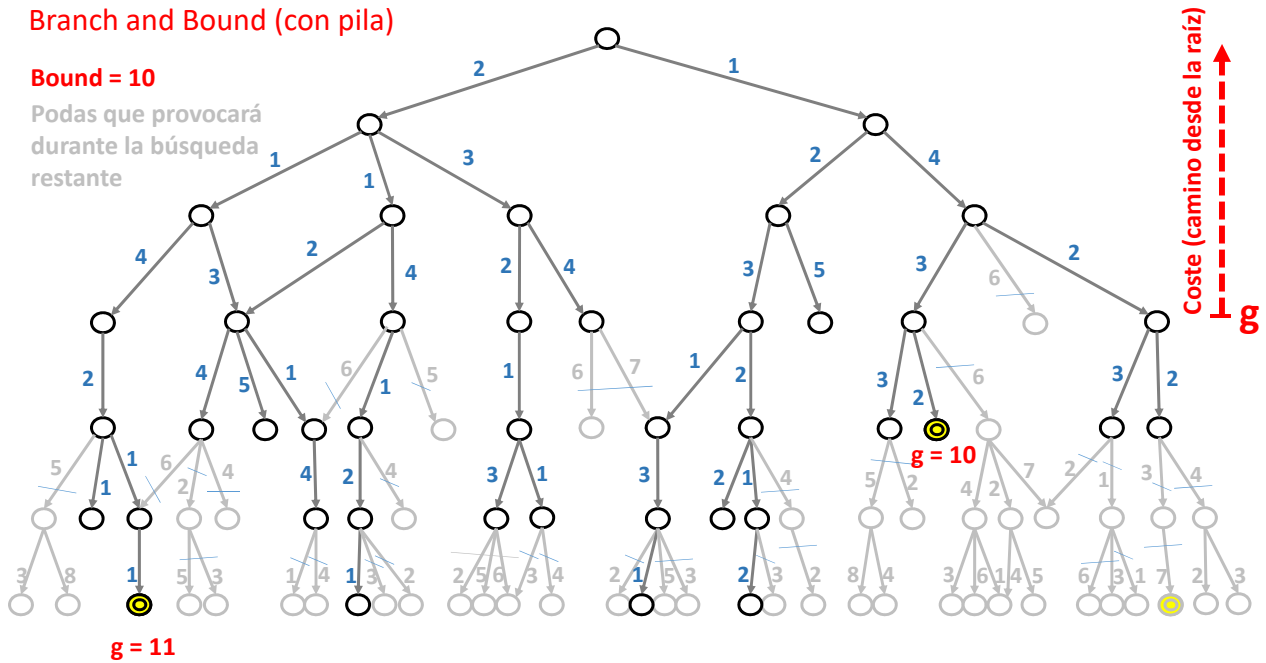
**Bound = 11**

El coste de la solución es menor que el bound → Bound = 10

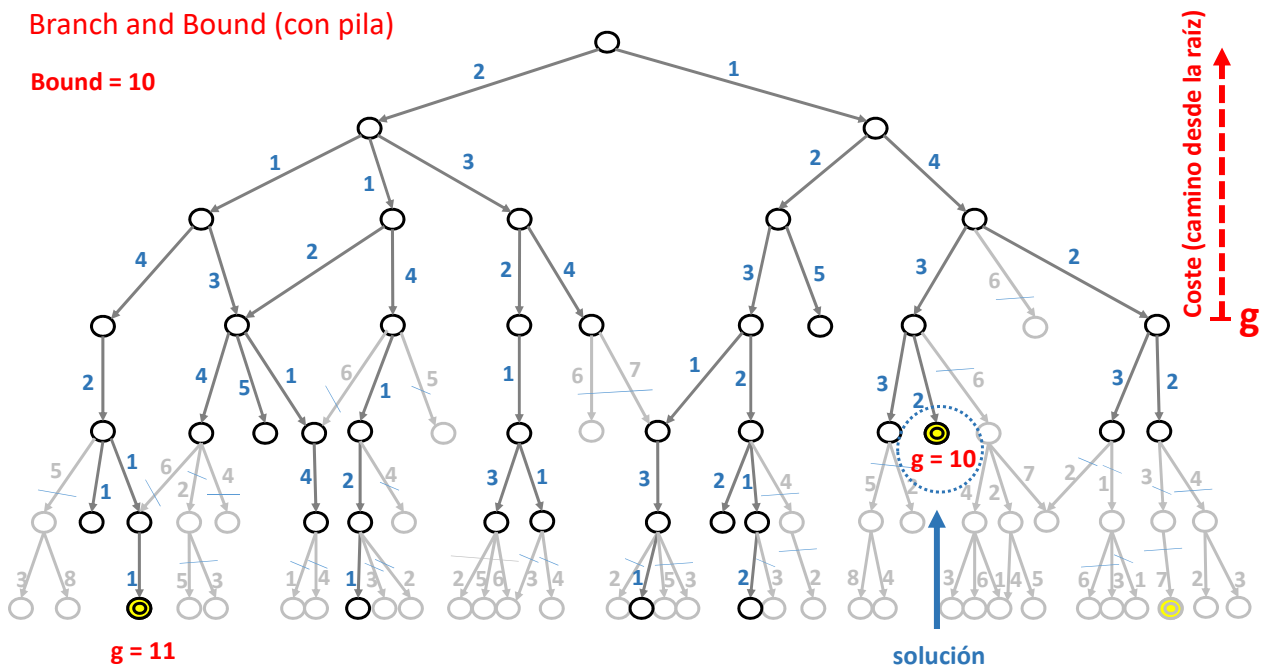
## Branch and Bound (con pila)

**Bound = 10**

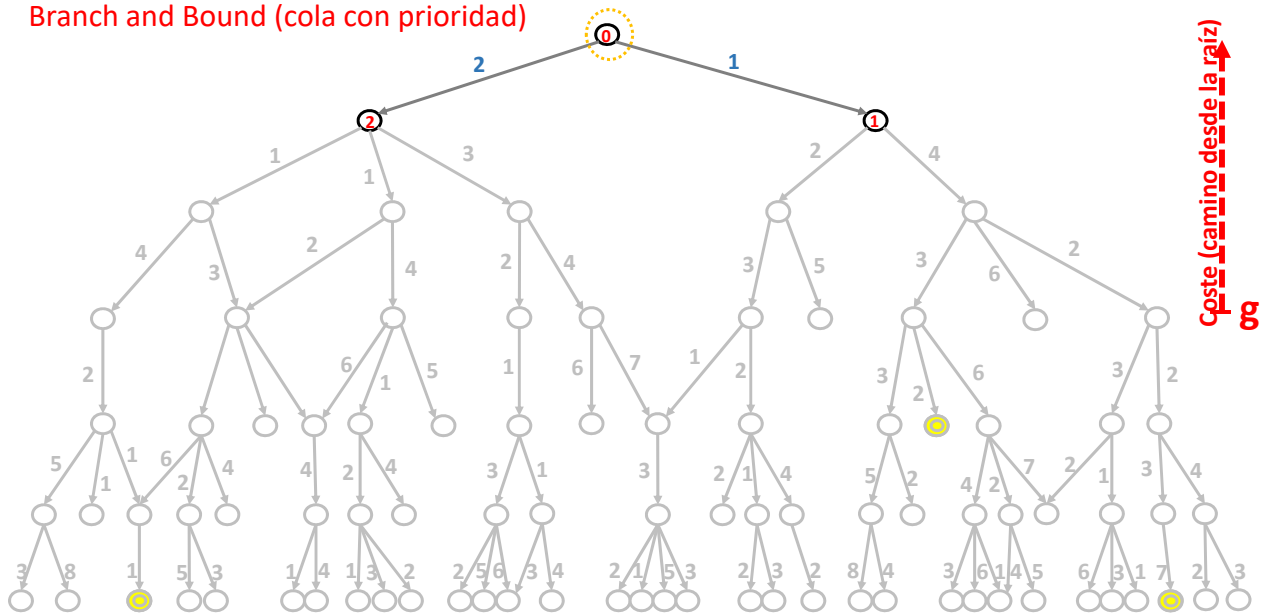
Podas que provocará durante la búsqueda restante



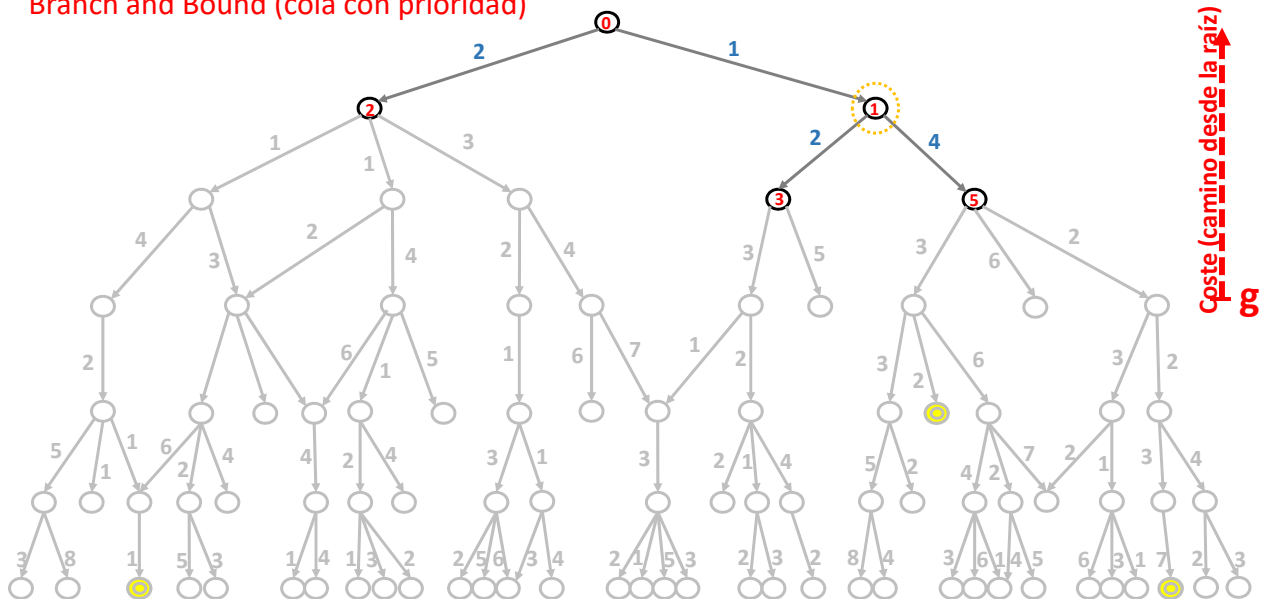
## Branch and Bound (con pila)

**Bound = 10**

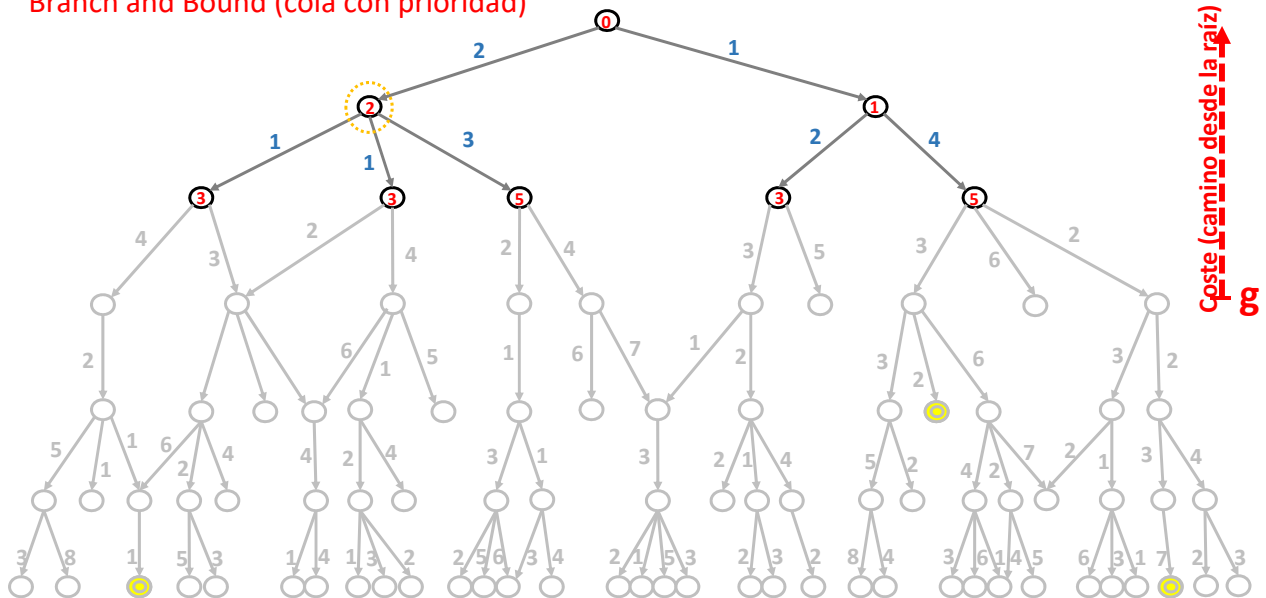
Branch and Bound (cola con prioridad)



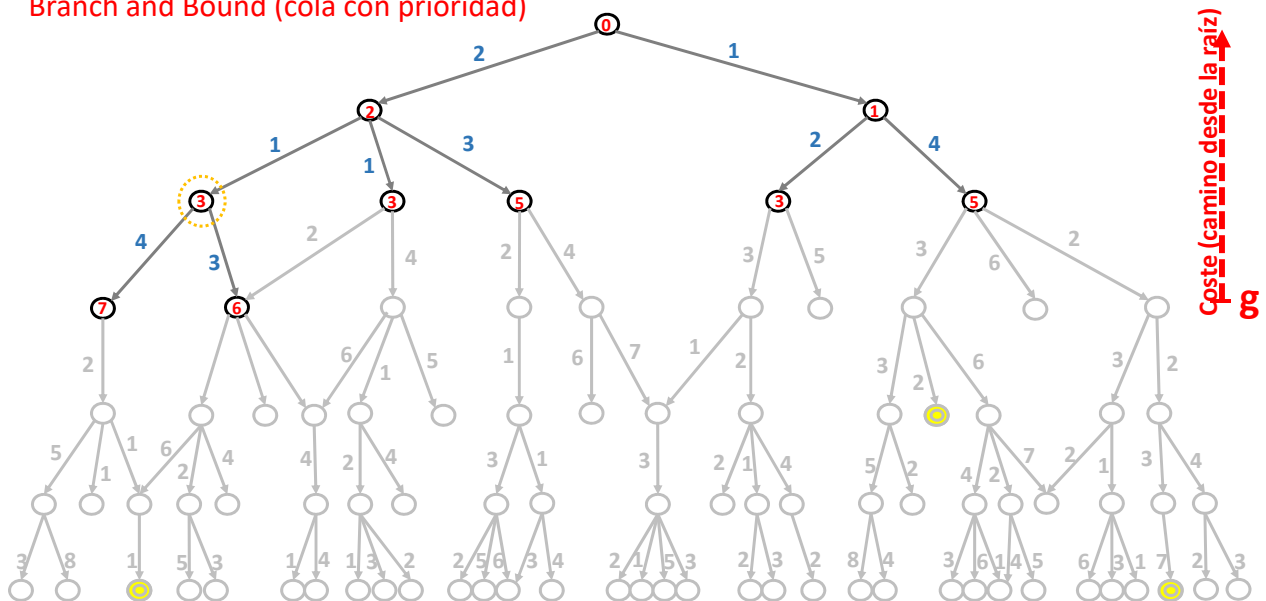
Branch and Bound (cola con prioridad)



### Branch and Bound (cola con prioridad)

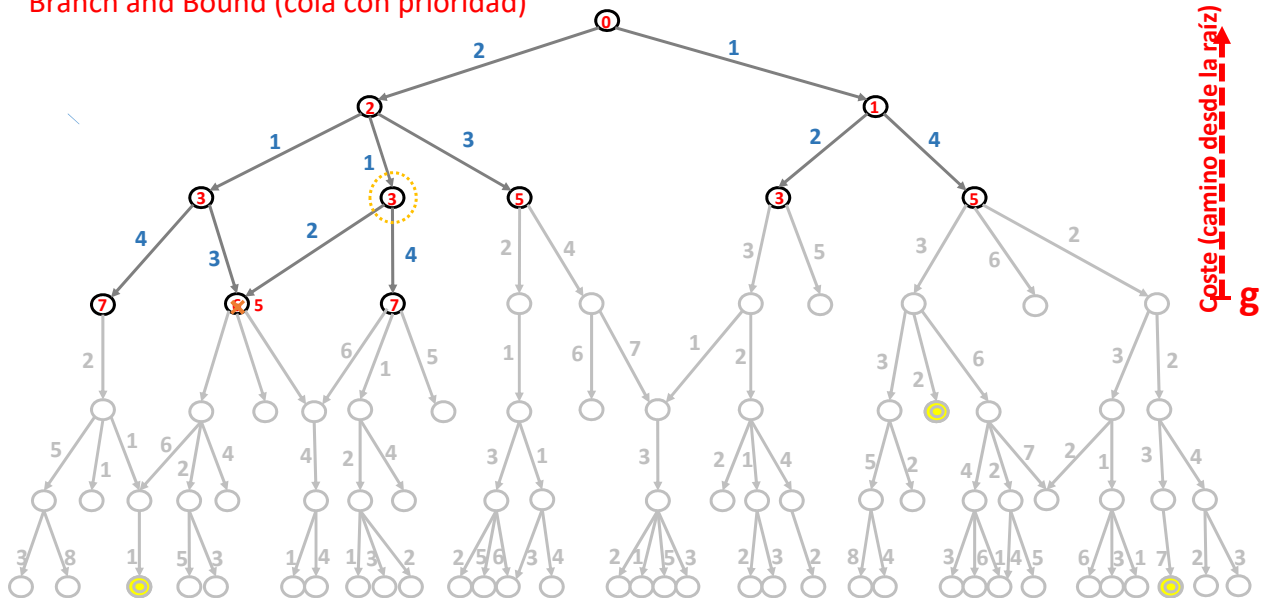


### Branch and Bound (cola con prioridad)

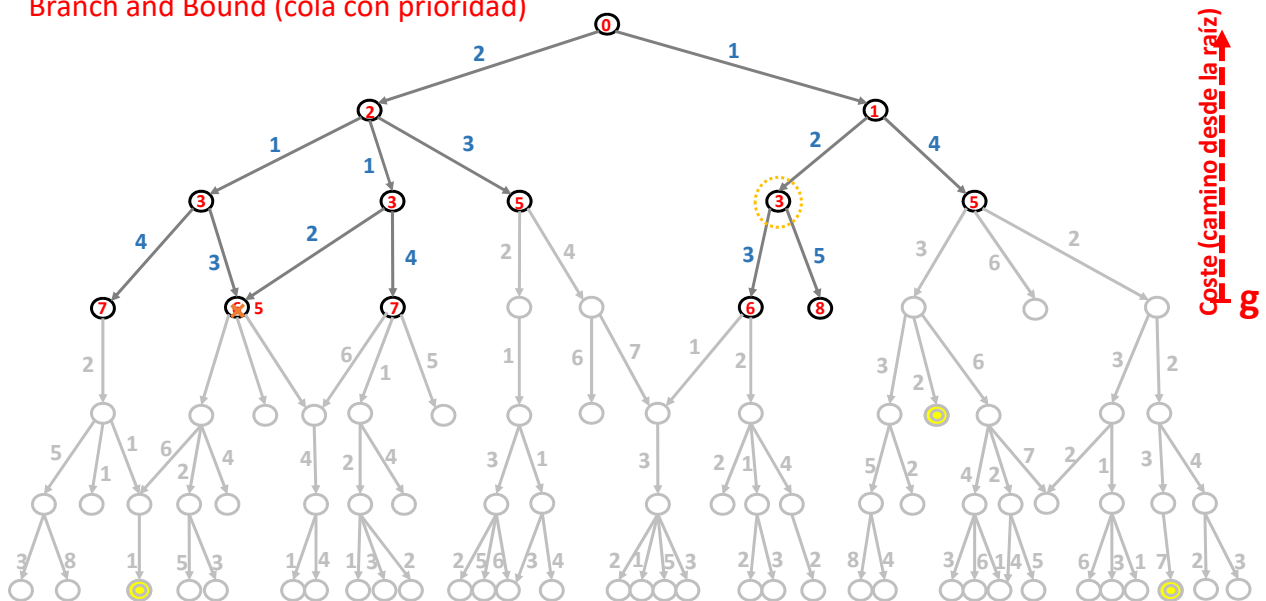




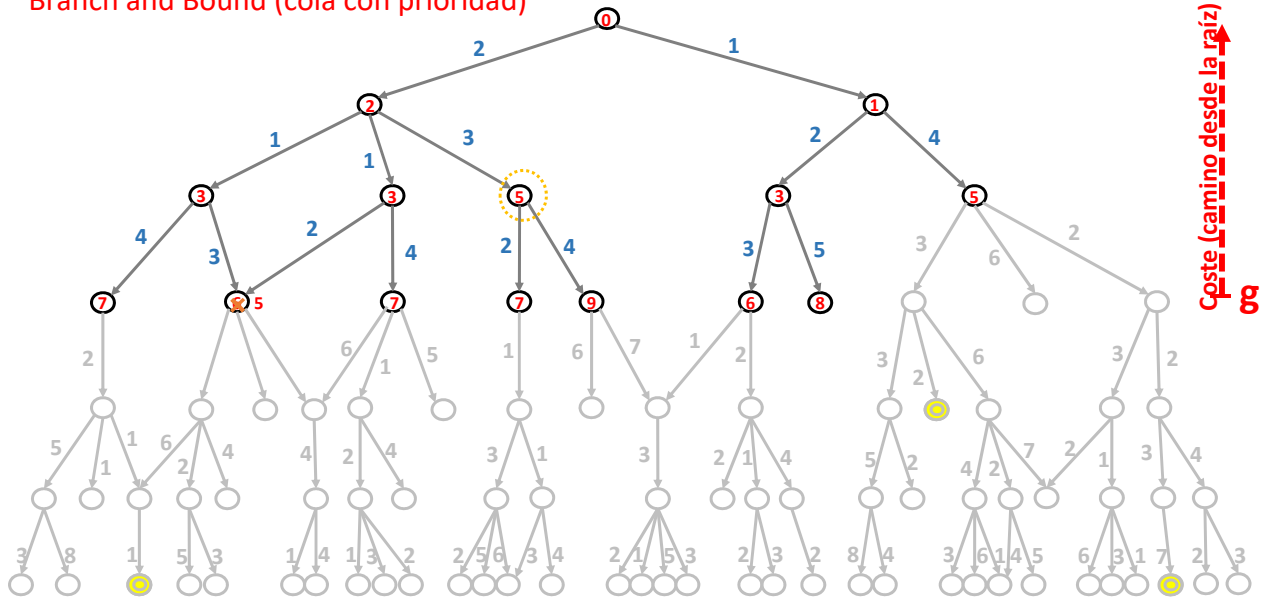
### Branch and Bound (cola con prioridad)



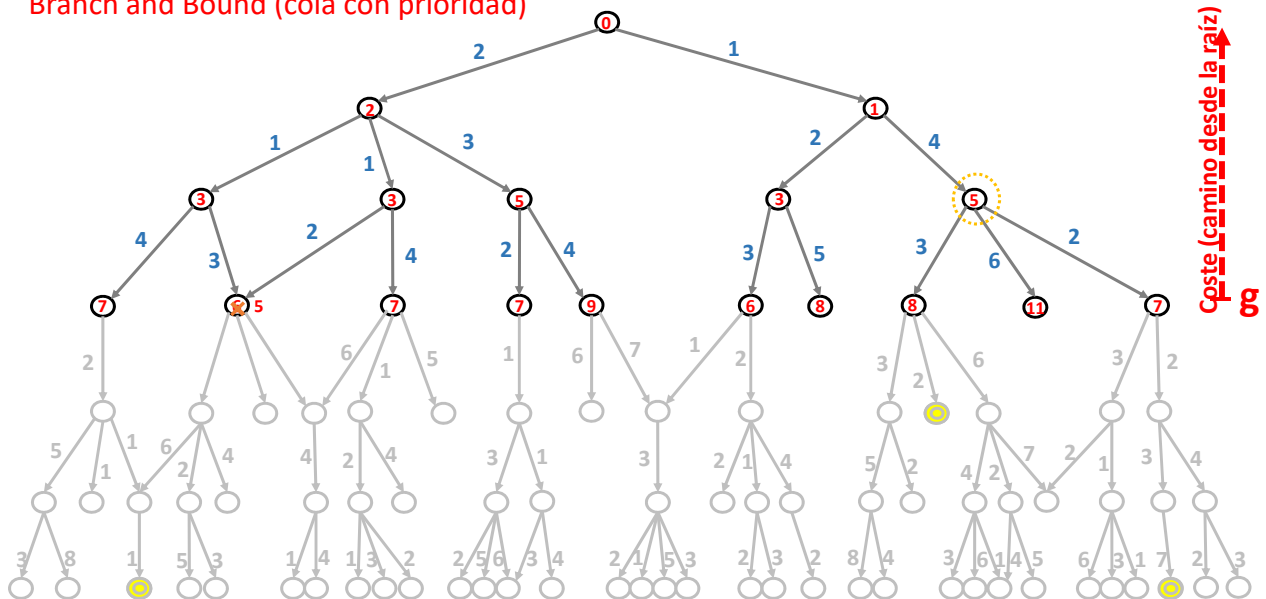
### Branch and Bound (cola con prioridad)



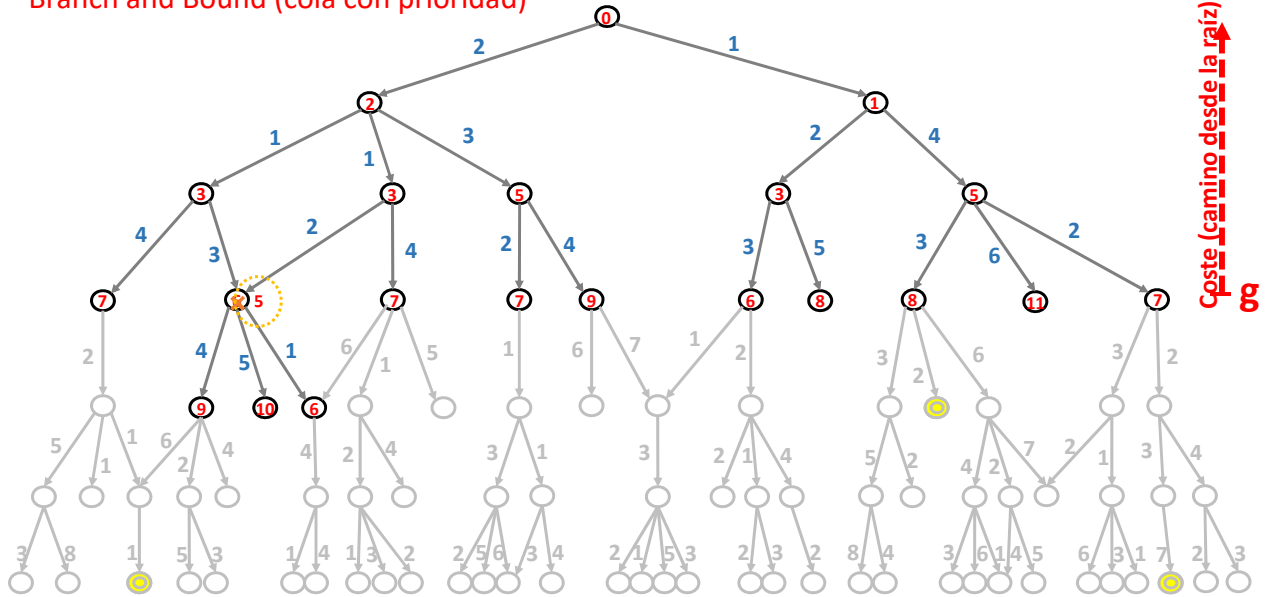
## Branch and Bound (cola con prioridad)



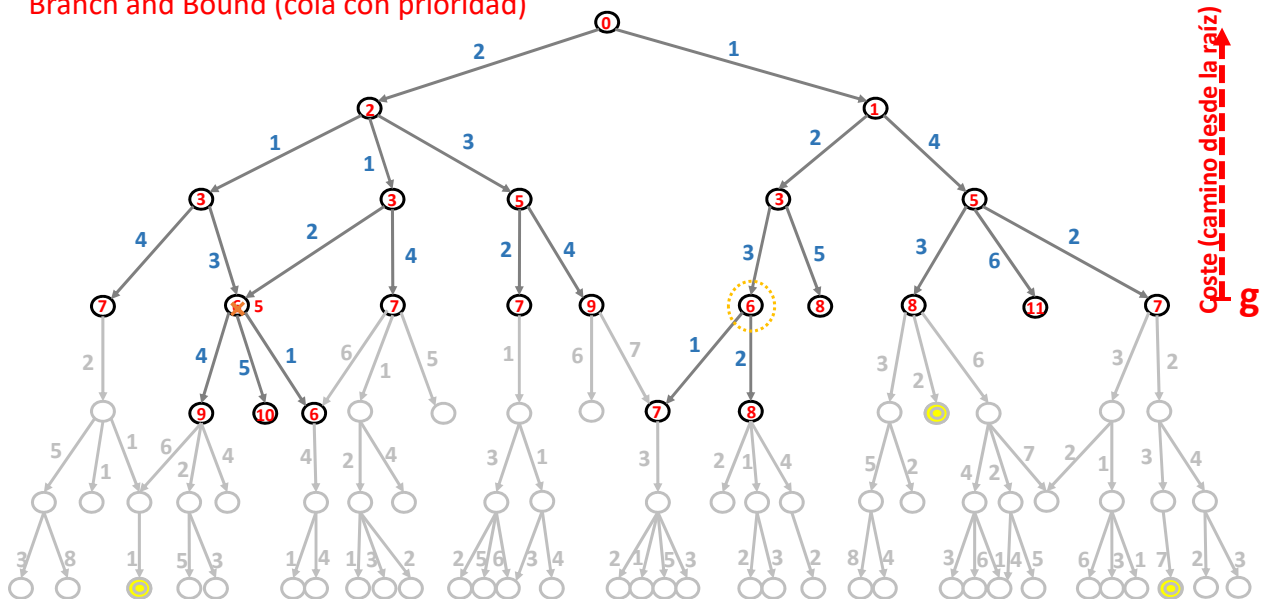
## Branch and Bound (cola con prioridad)



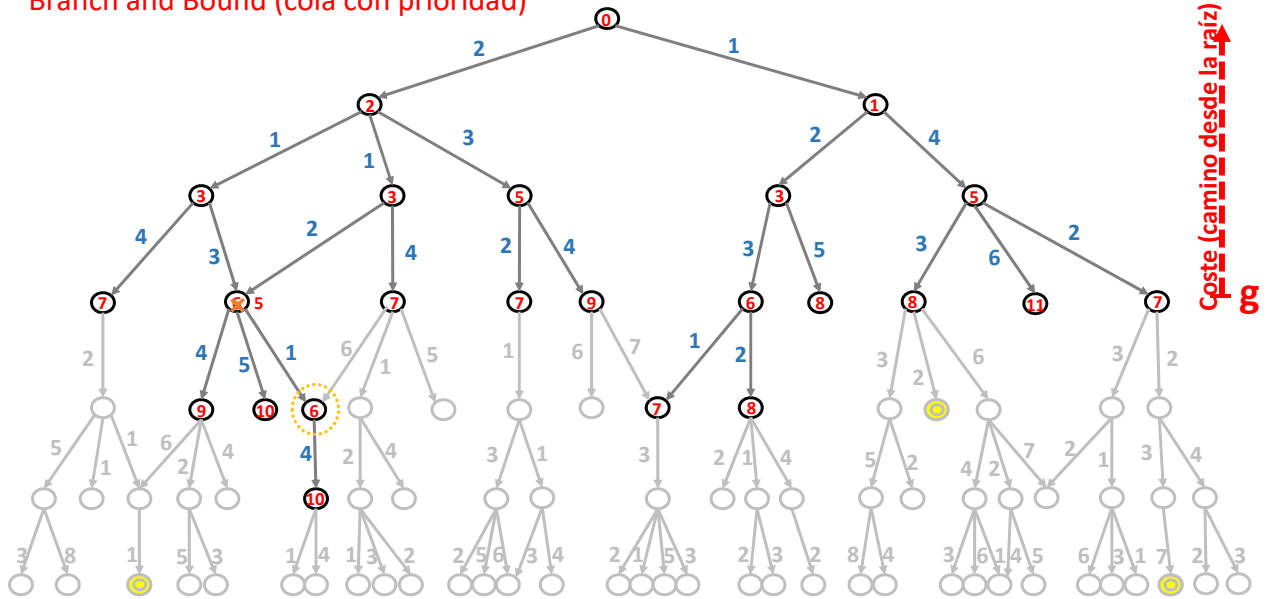
## Branch and Bound (cola con prioridad)



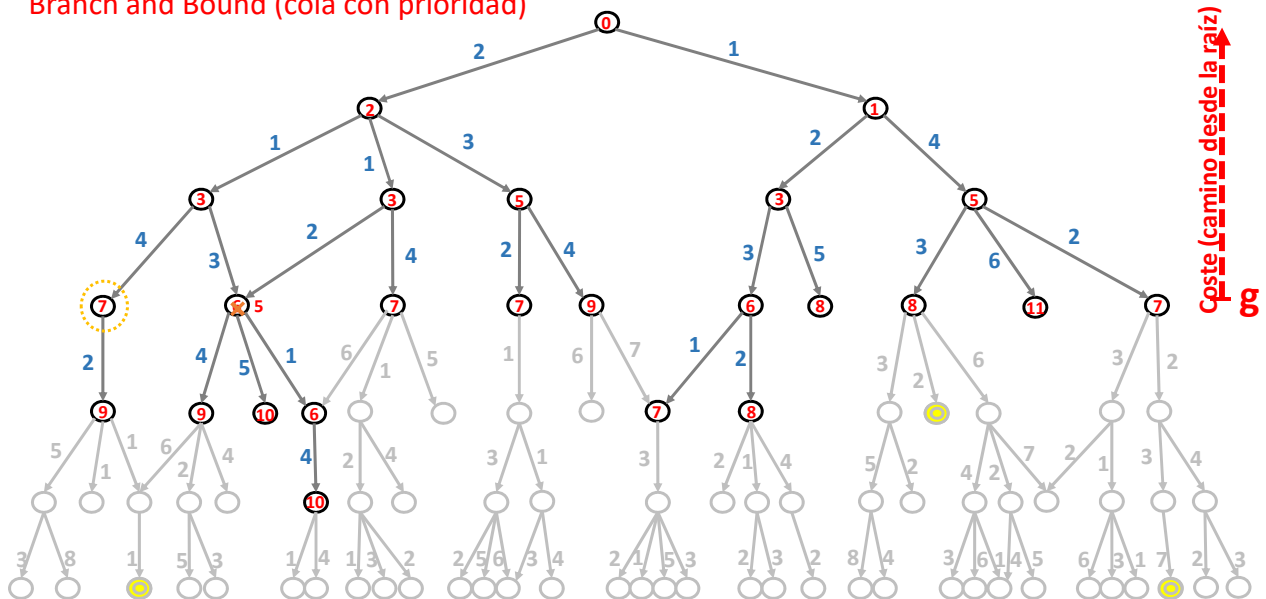
## Branch and Bound (cola con prioridad)



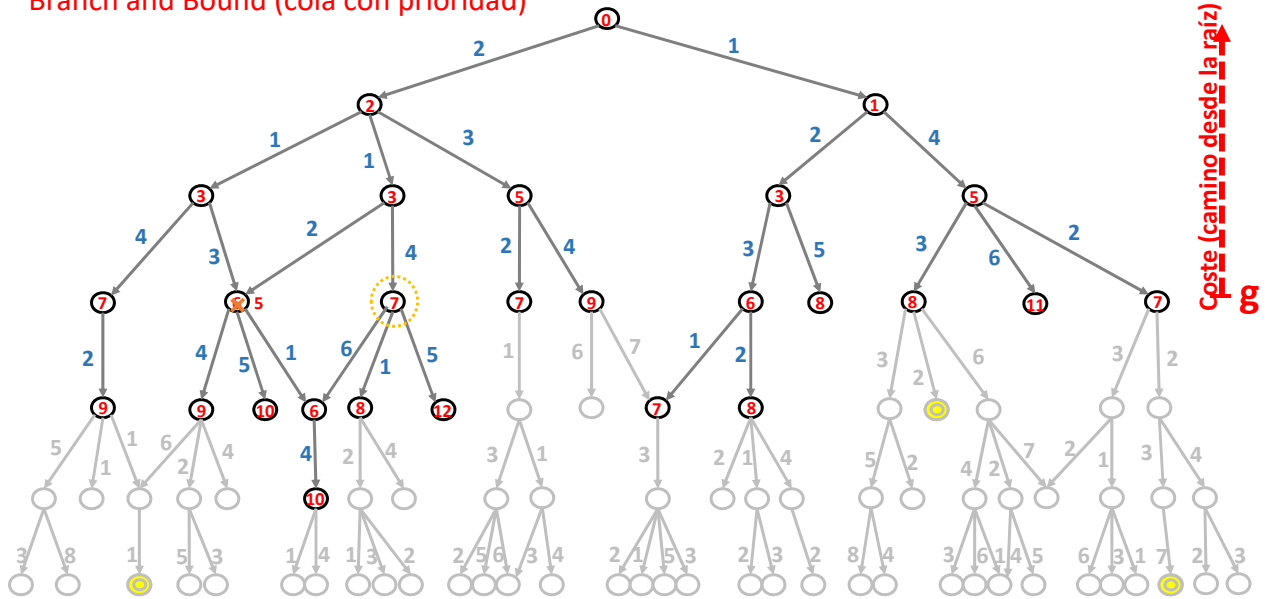
Branch and Bound (cola con prioridad)



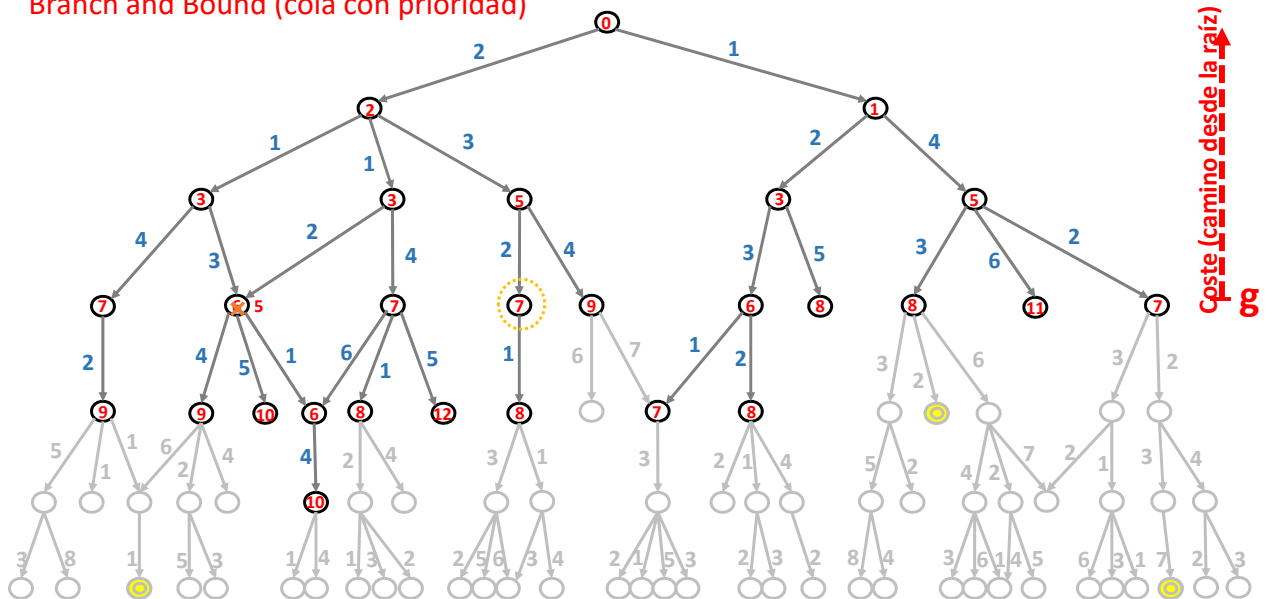
Branch and Bound (cola con prioridad)



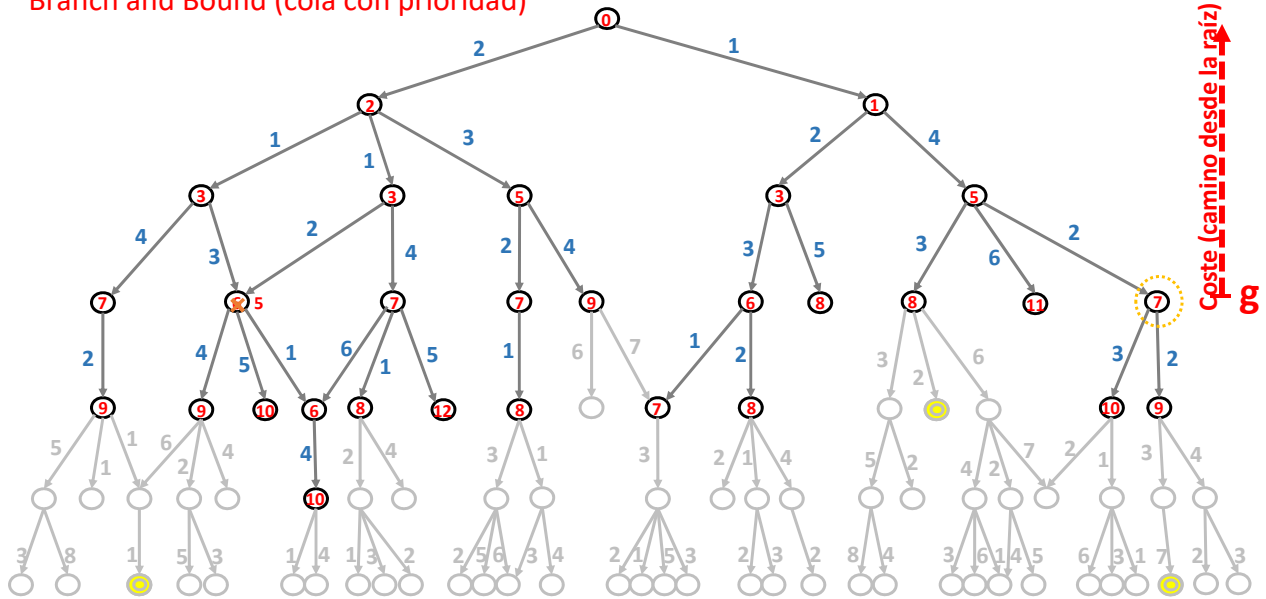
## Branch and Bound (cola con prioridad)



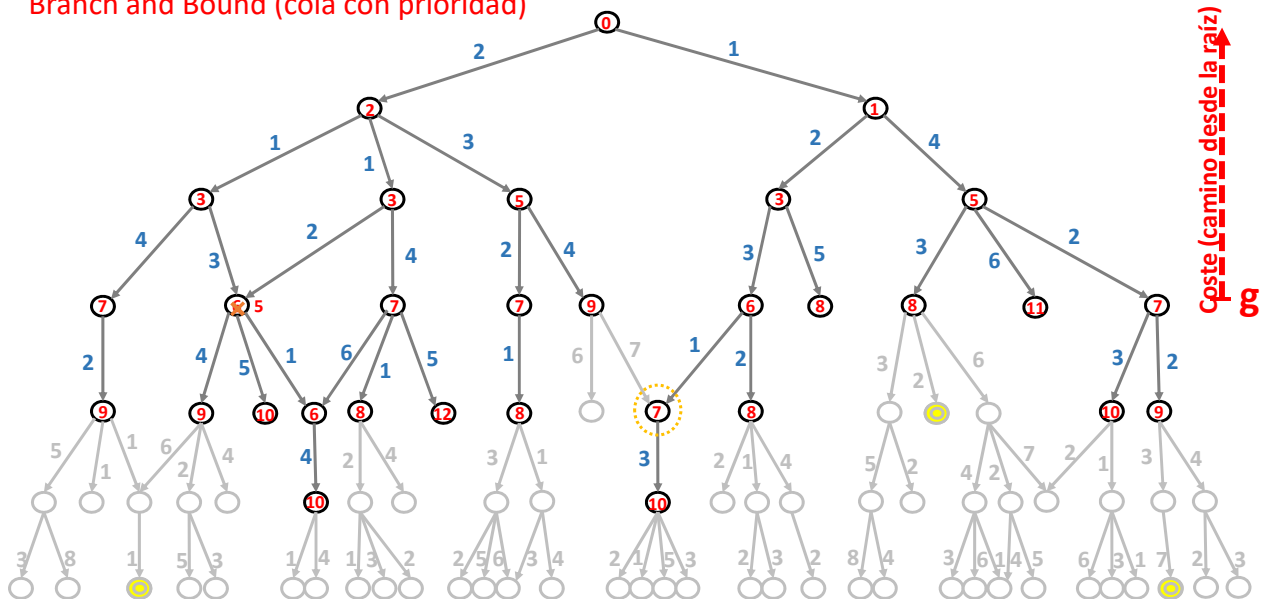
## Branch and Bound (cola con prioridad)



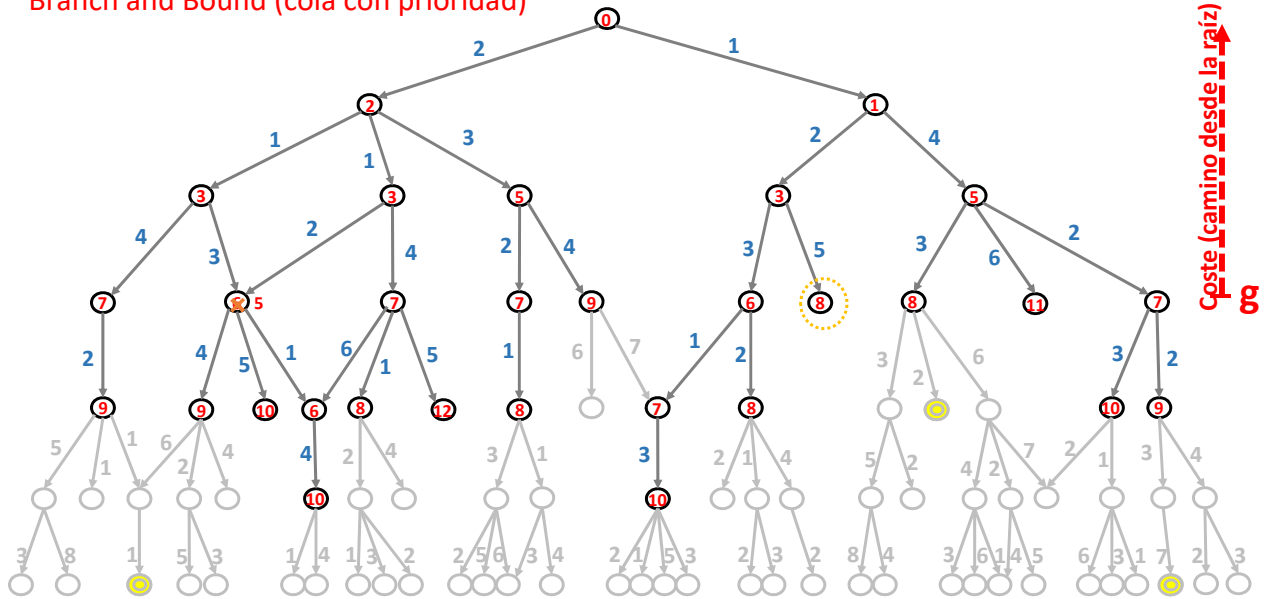
## Branch and Bound (cola con prioridad)



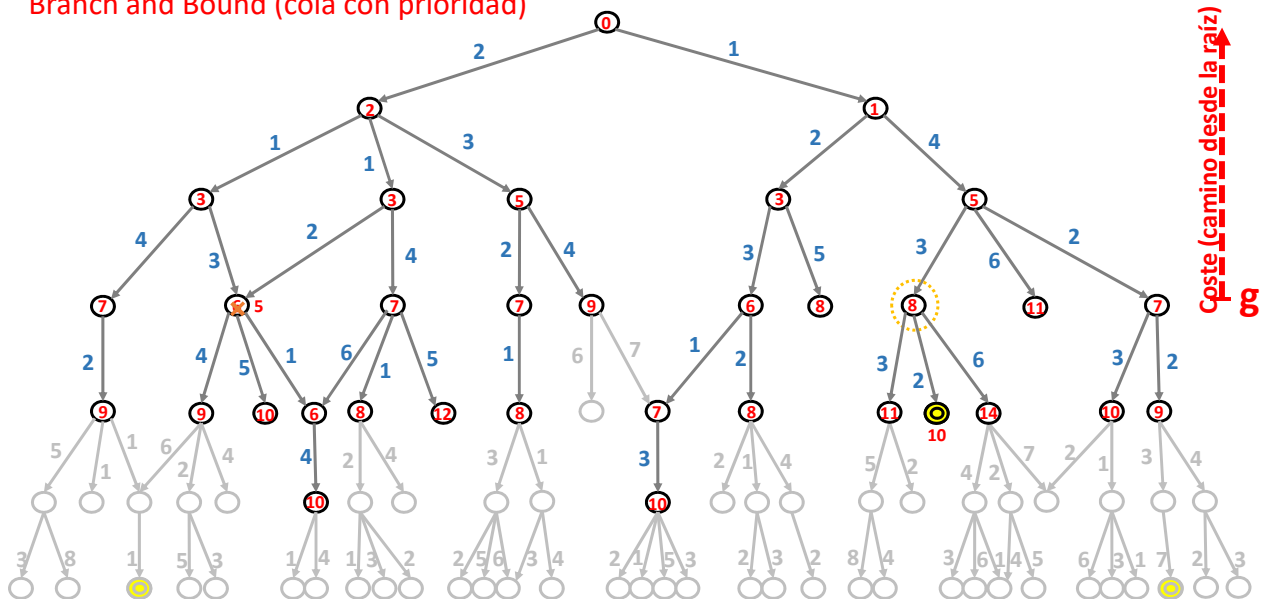
## Branch and Bound (cola con prioridad)



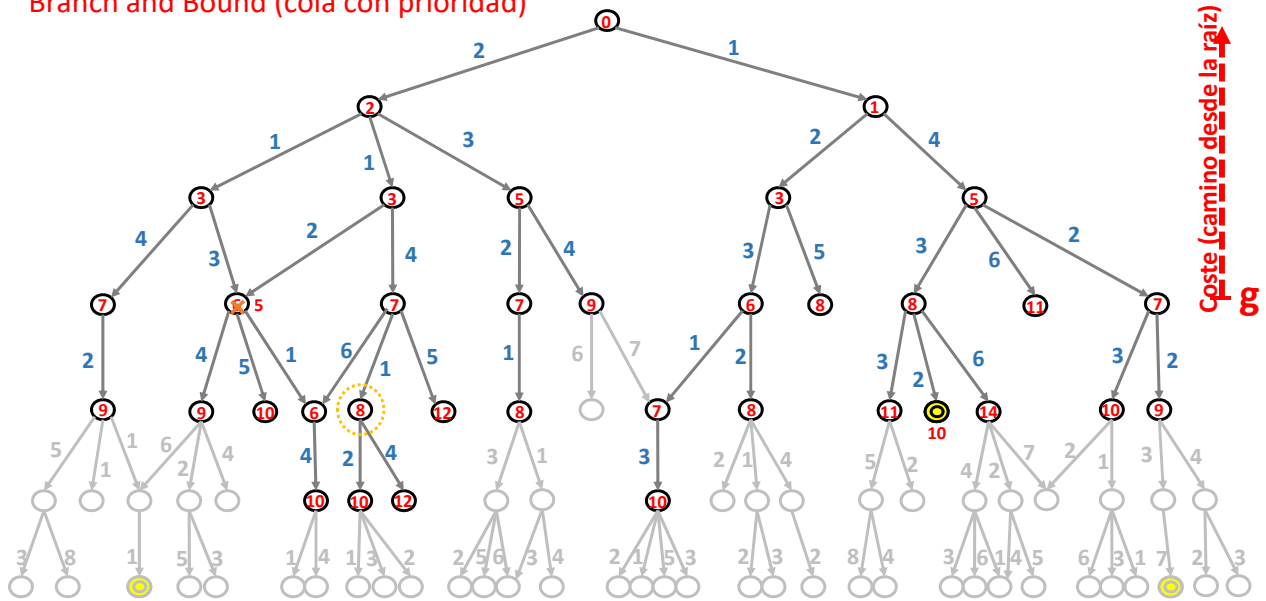
## Branch and Bound (cola con prioridad)



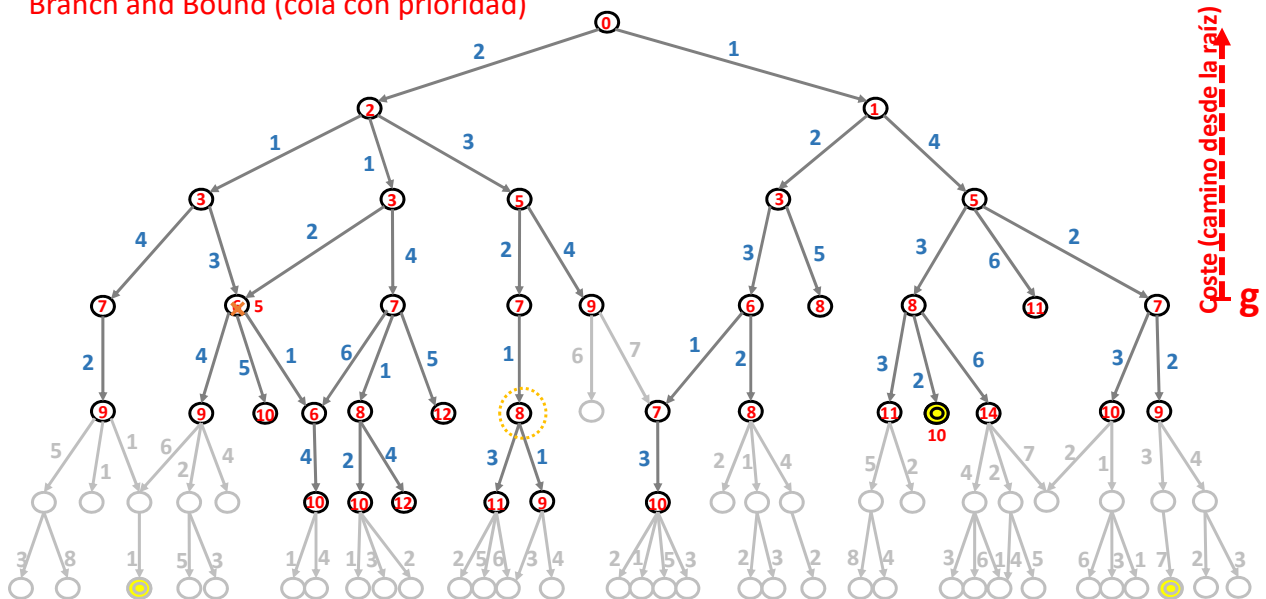
## Branch and Bound (cola con prioridad)



## Branch and Bound (cola con prioridad)

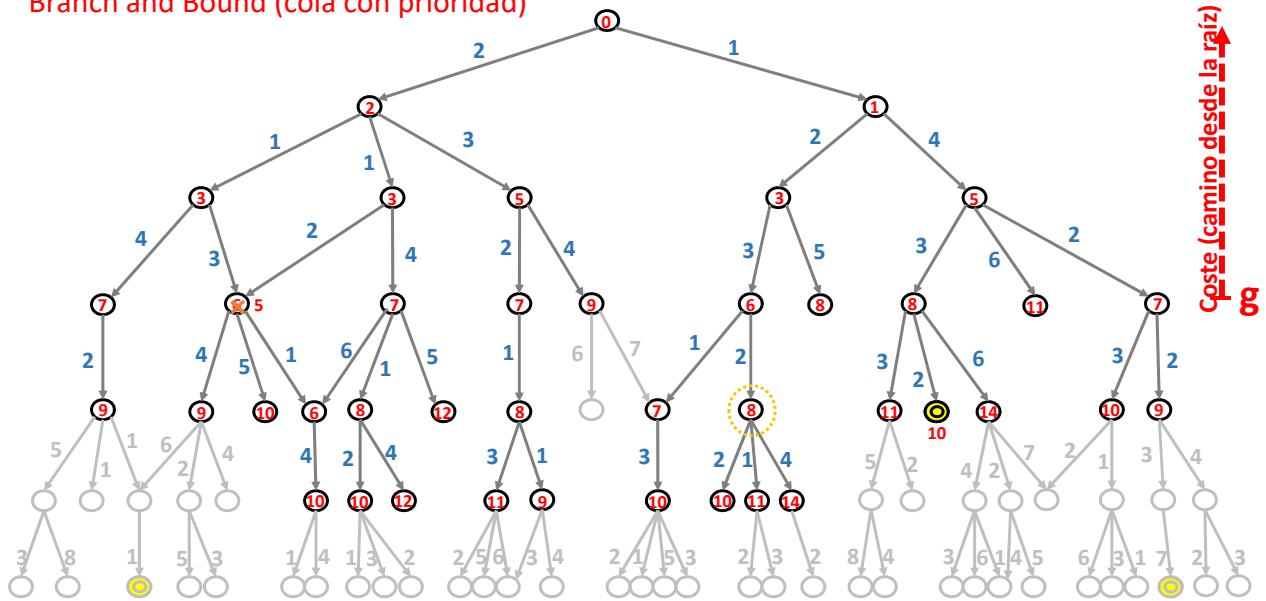


## Branch and Bound (cola con prioridad)

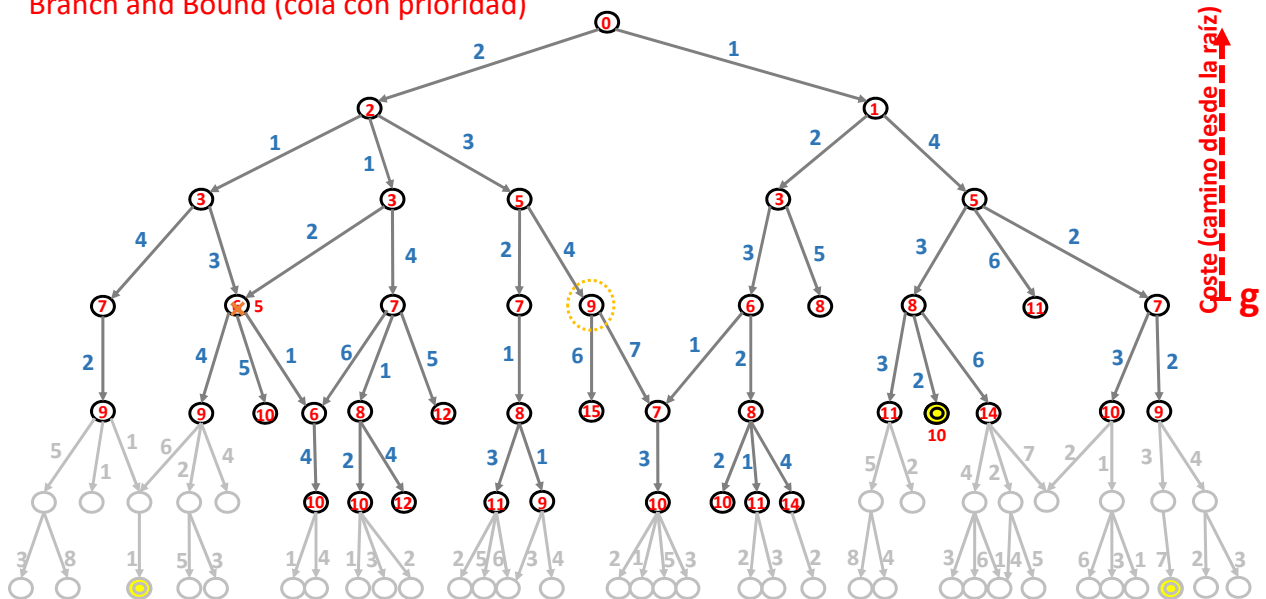




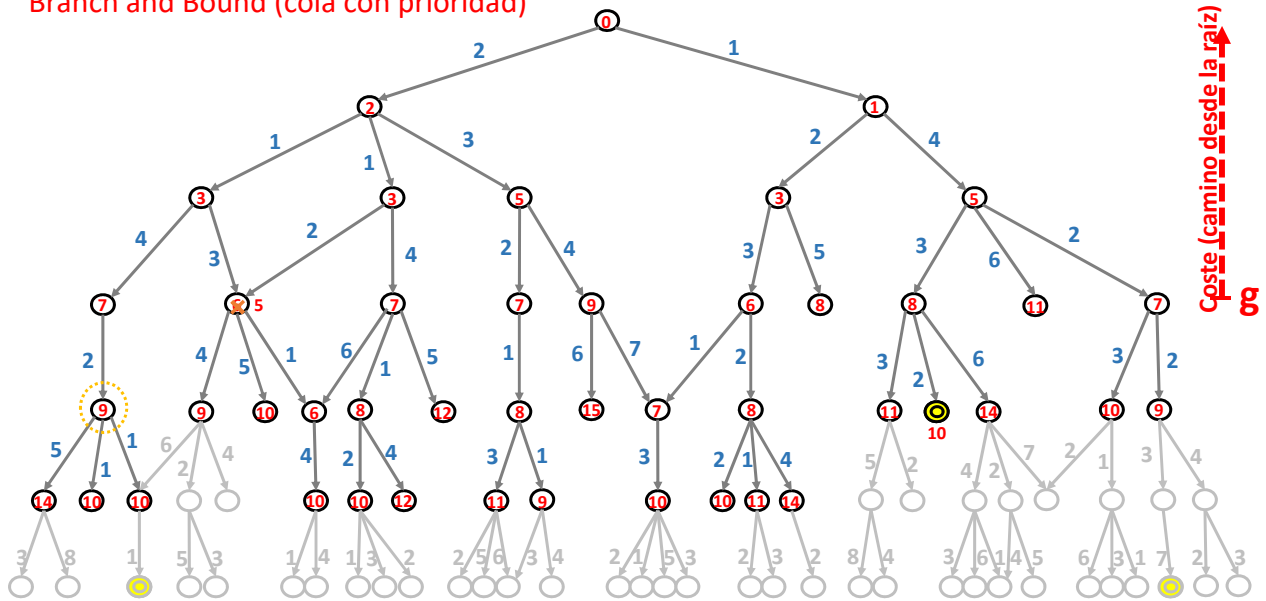
## Branch and Bound (cola con prioridad)



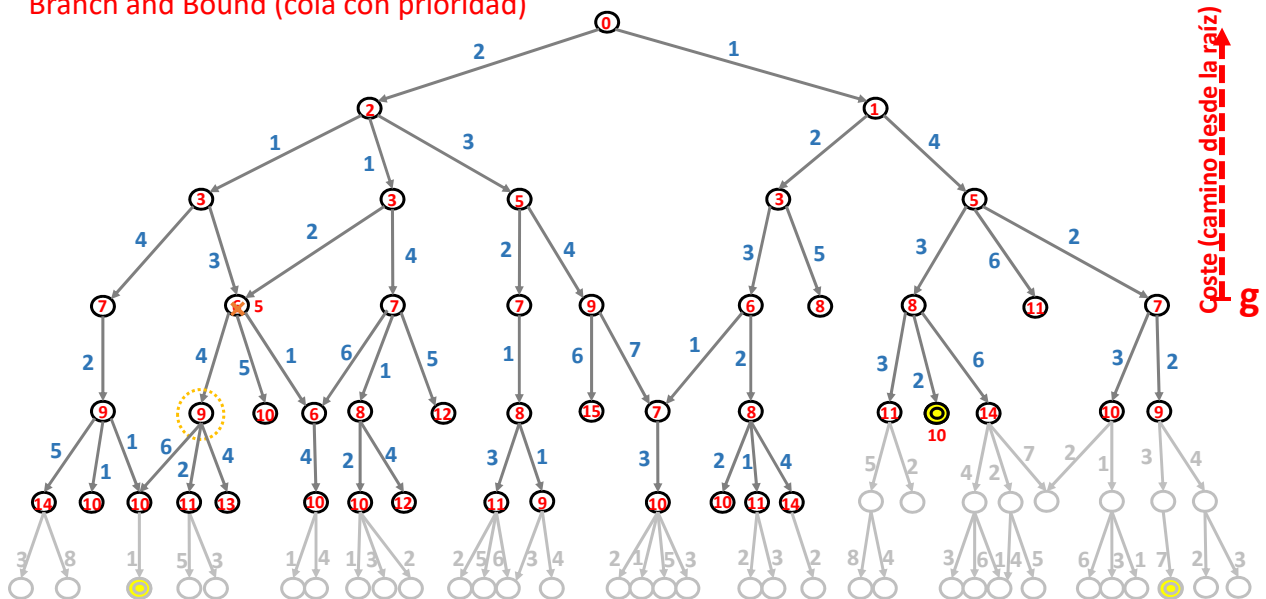
## Branch and Bound (cola con prioridad)



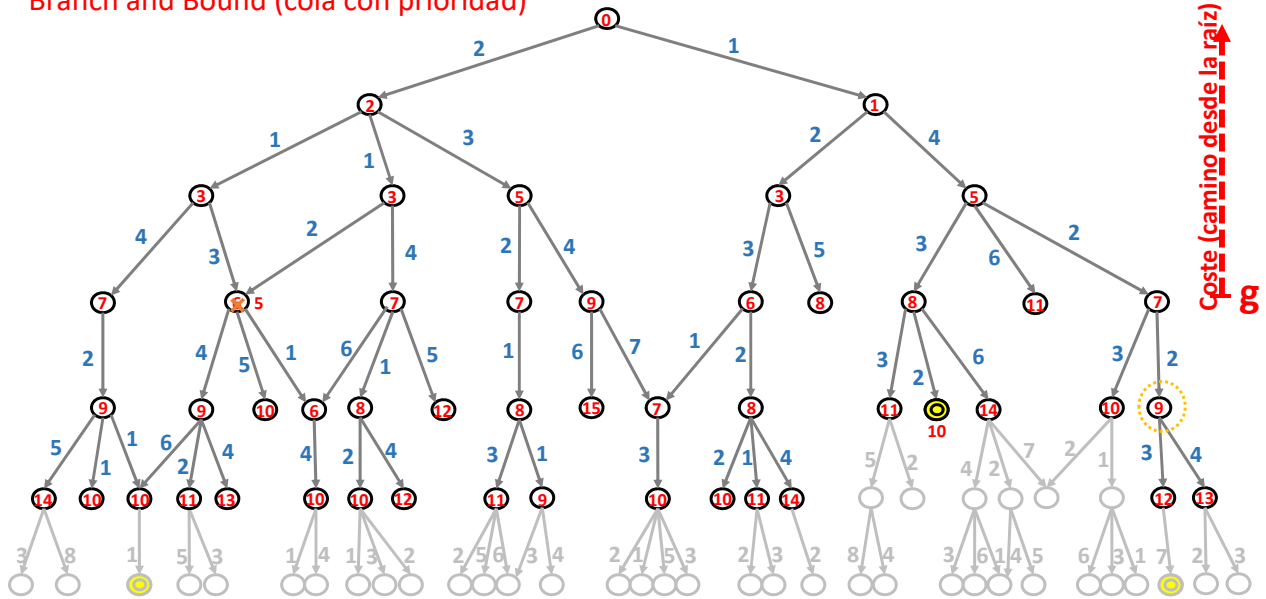
## Branch and Bound (cola con prioridad)



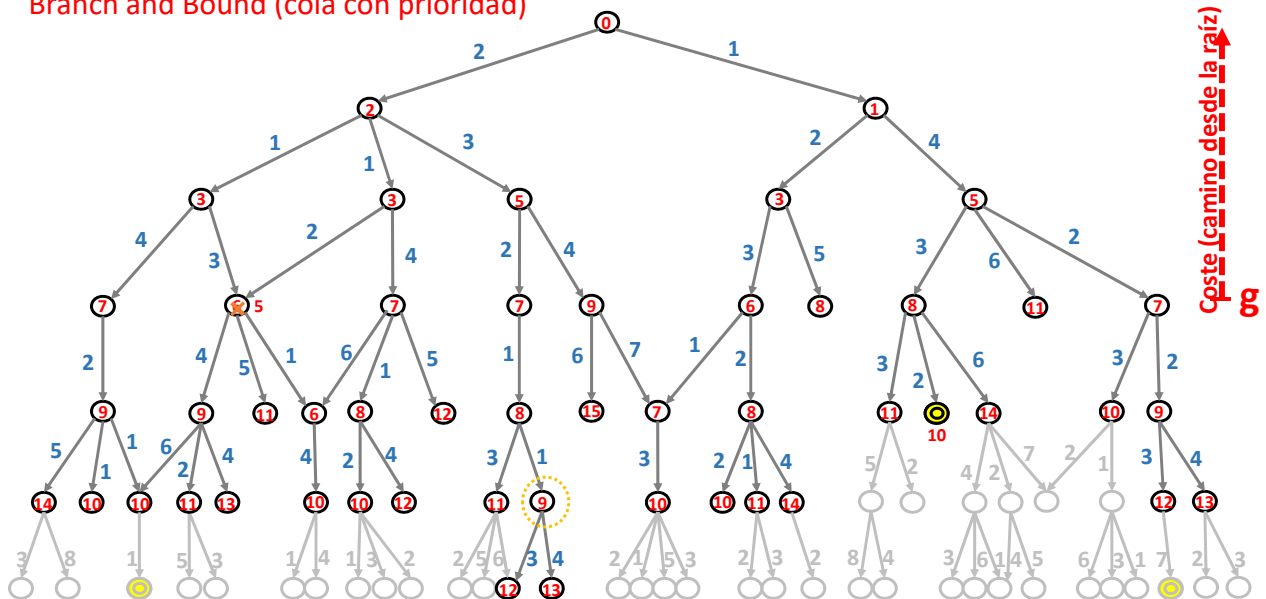
## Branch and Bound (cola con prioridad)



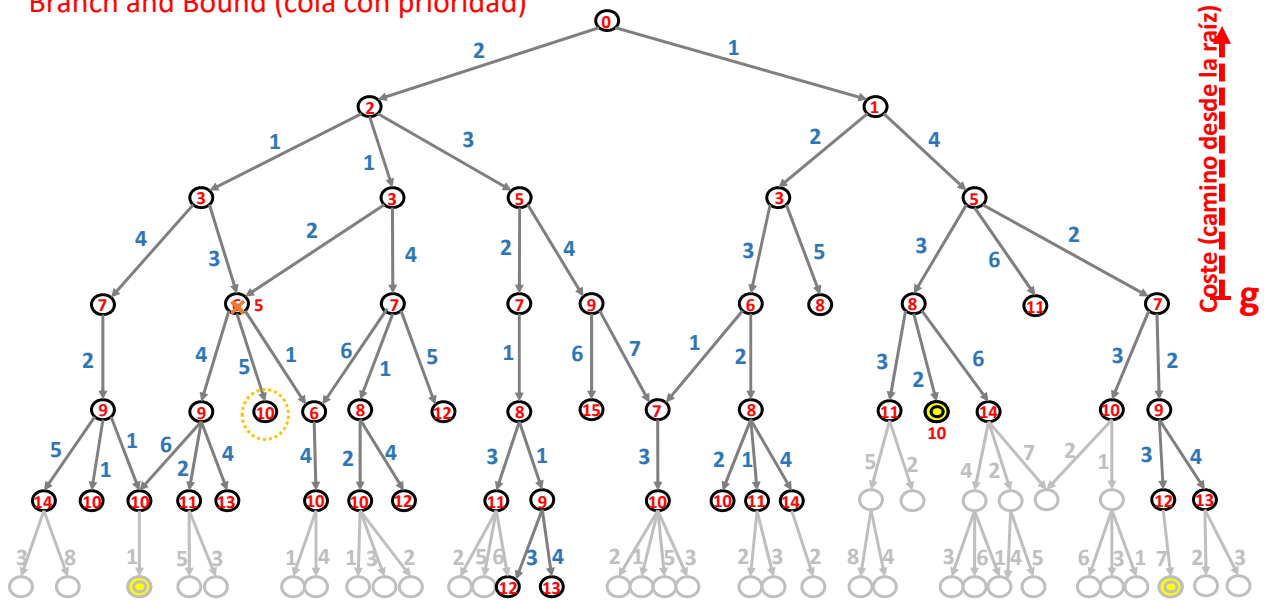
## Branch and Bound (cola con prioridad)



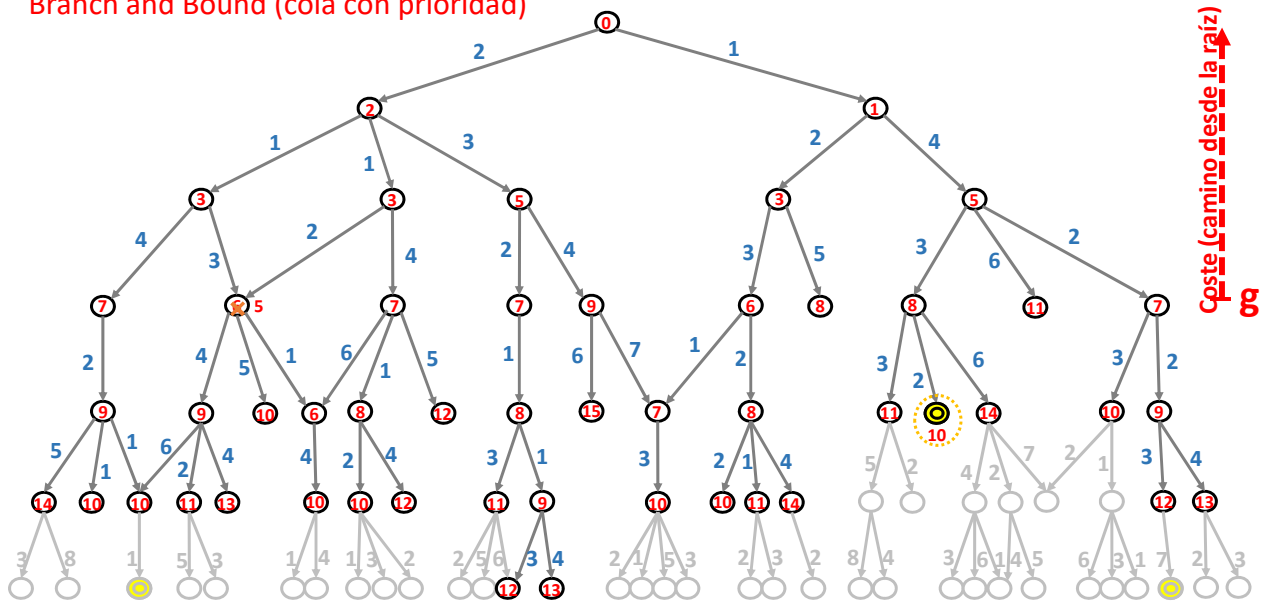
## Branch and Bound (cola con prioridad)



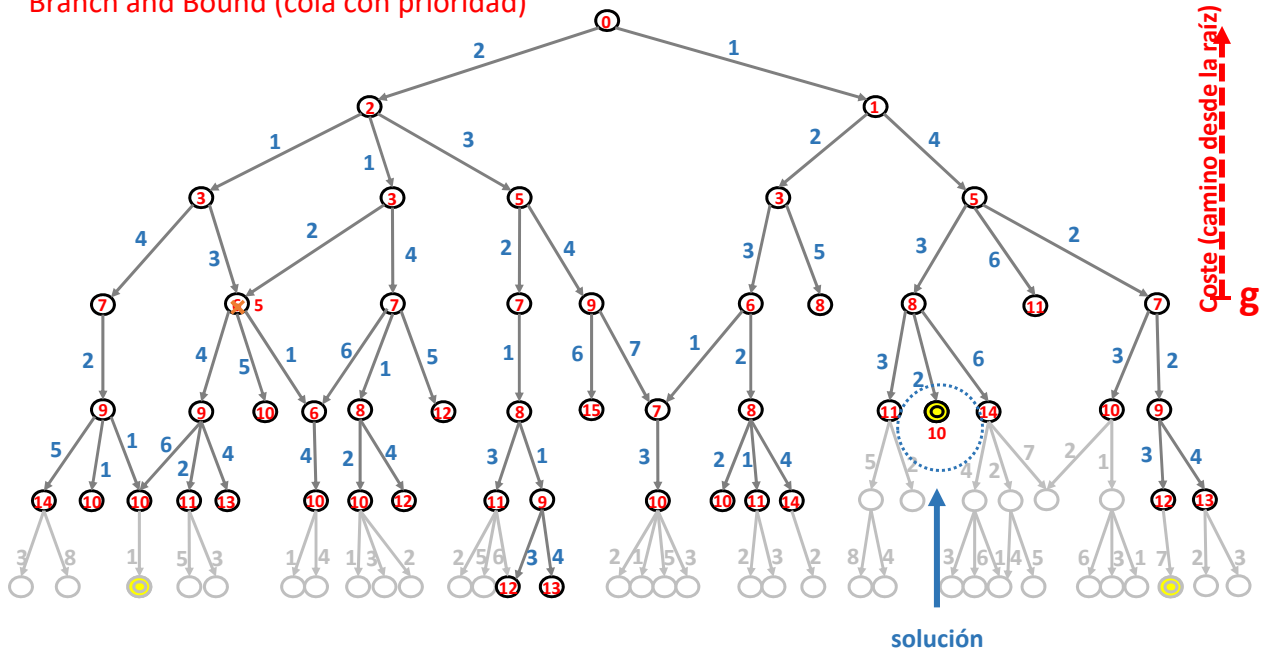
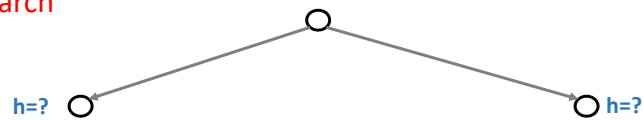
## Branch and Bound (cola con prioridad)



## Branch and Bound (cola con prioridad)



## Branch and Bound (cola con prioridad)

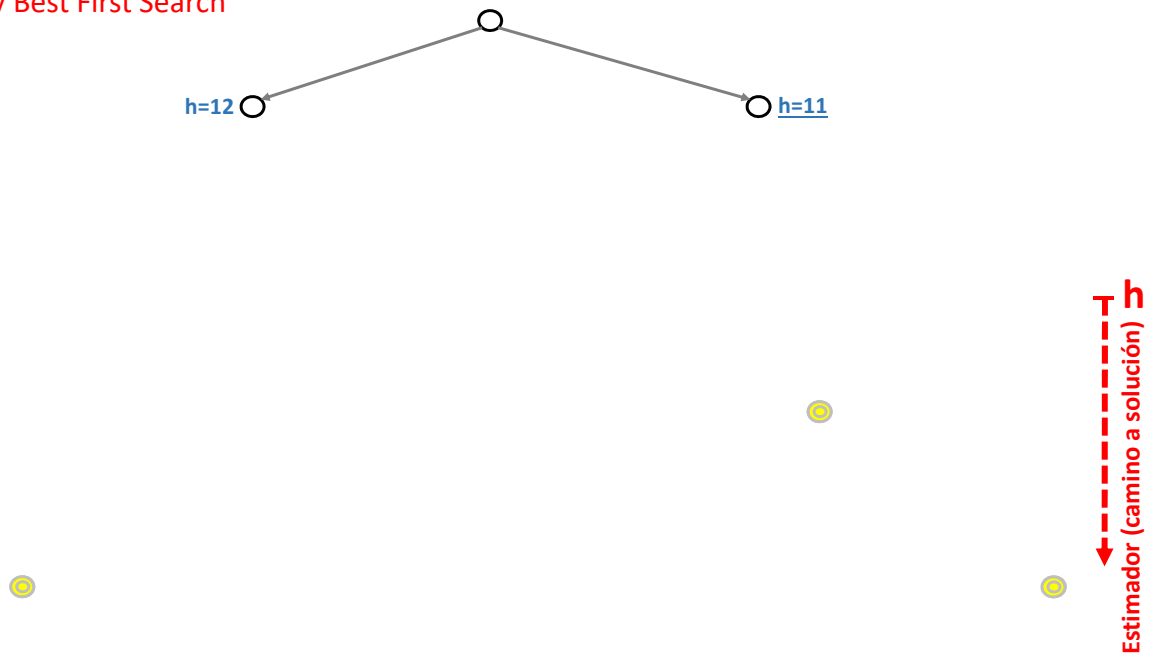
Greedy Best First Search  
(busqueda guiada  
por proximidad de  
la solución)

¿Por qué camino hemos de ir para encontrar una buena solución?

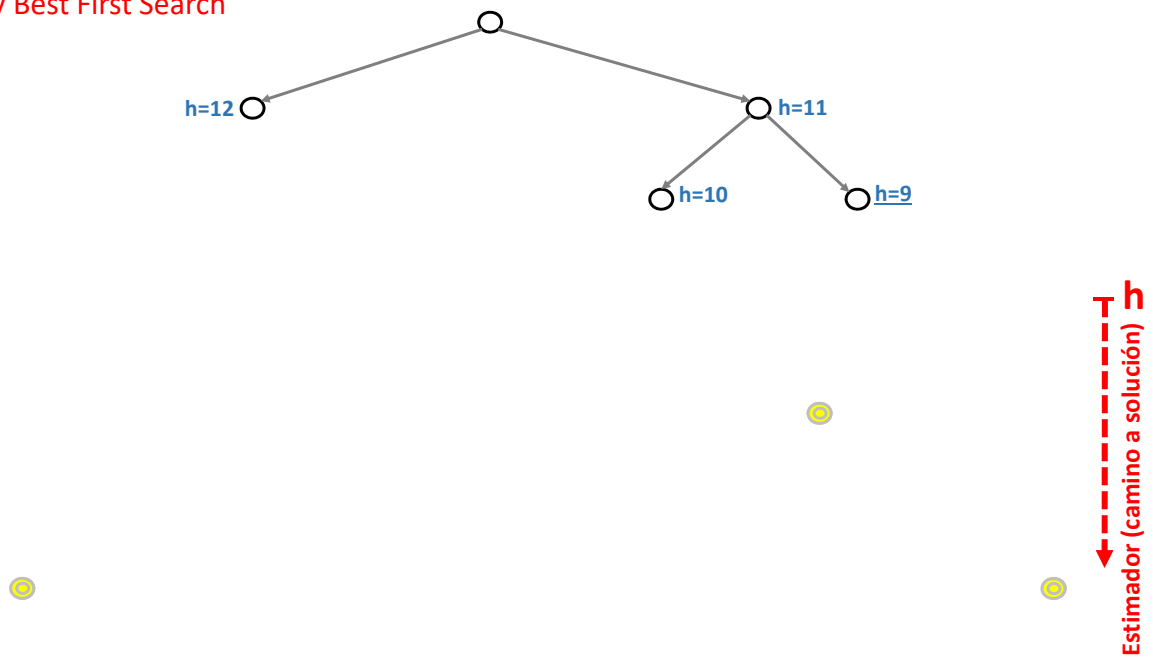
Estimador  $h$ , en este caso escogemos  $h$  menor



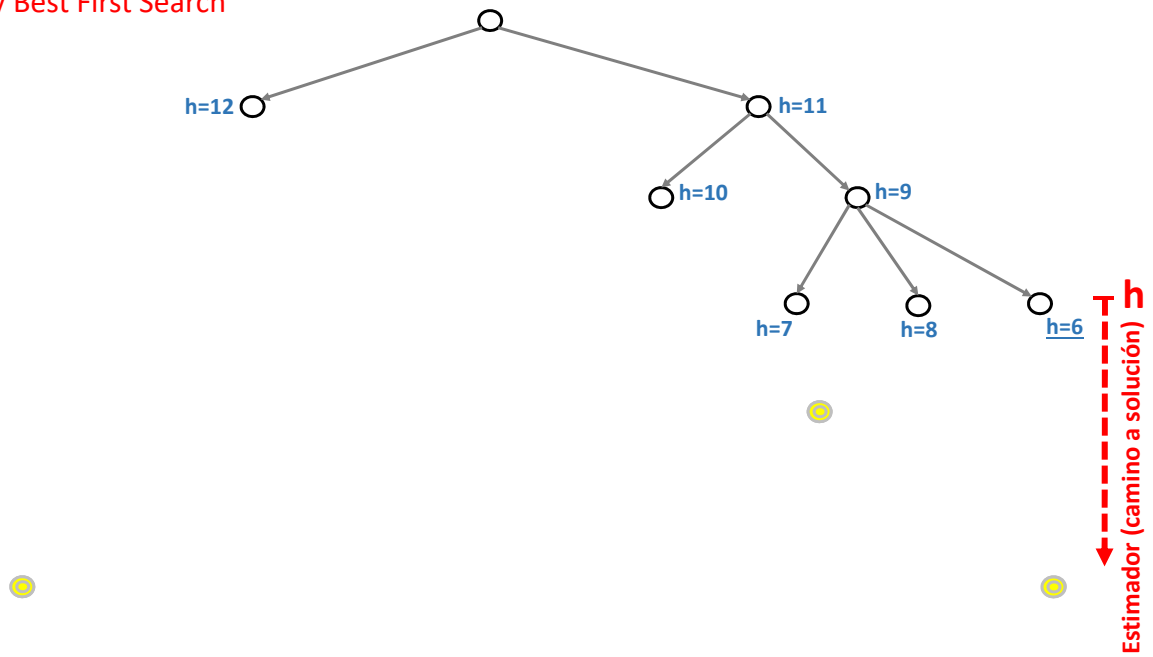
## Greedy Best First Search



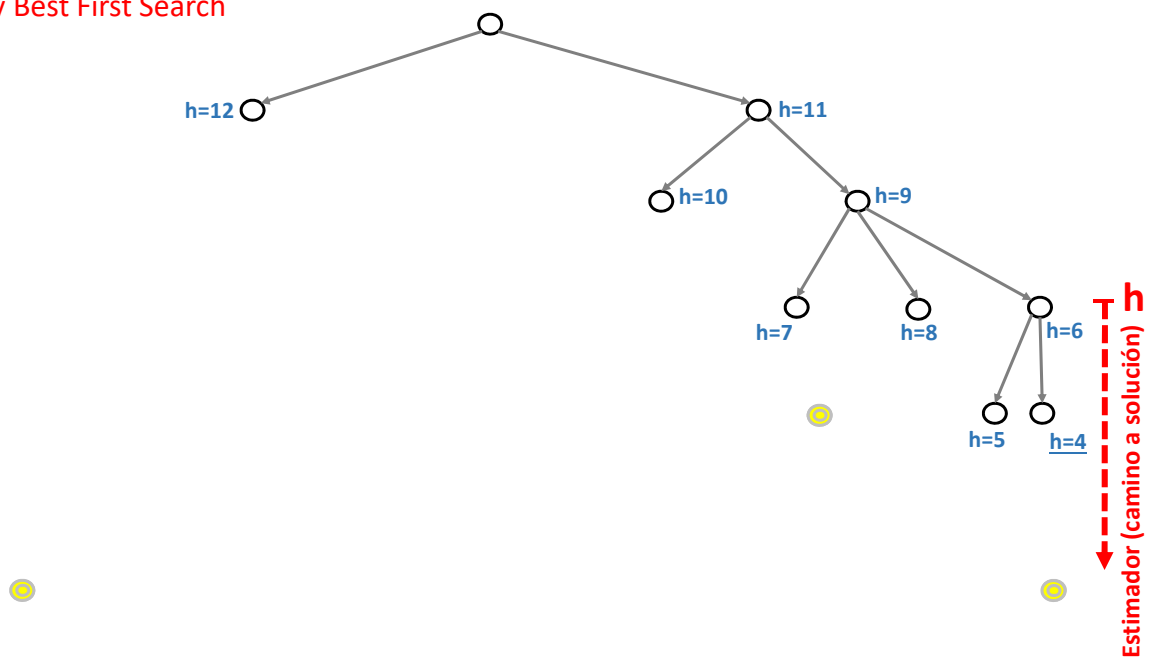
## Greedy Best First Search



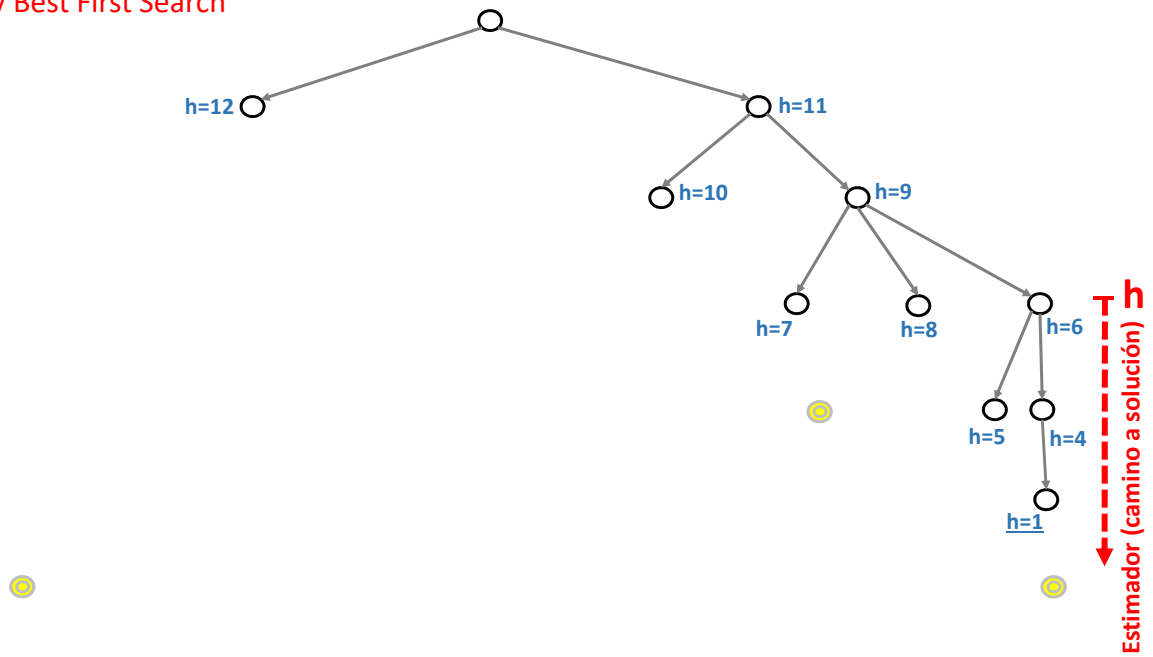
## Greedy Best First Search



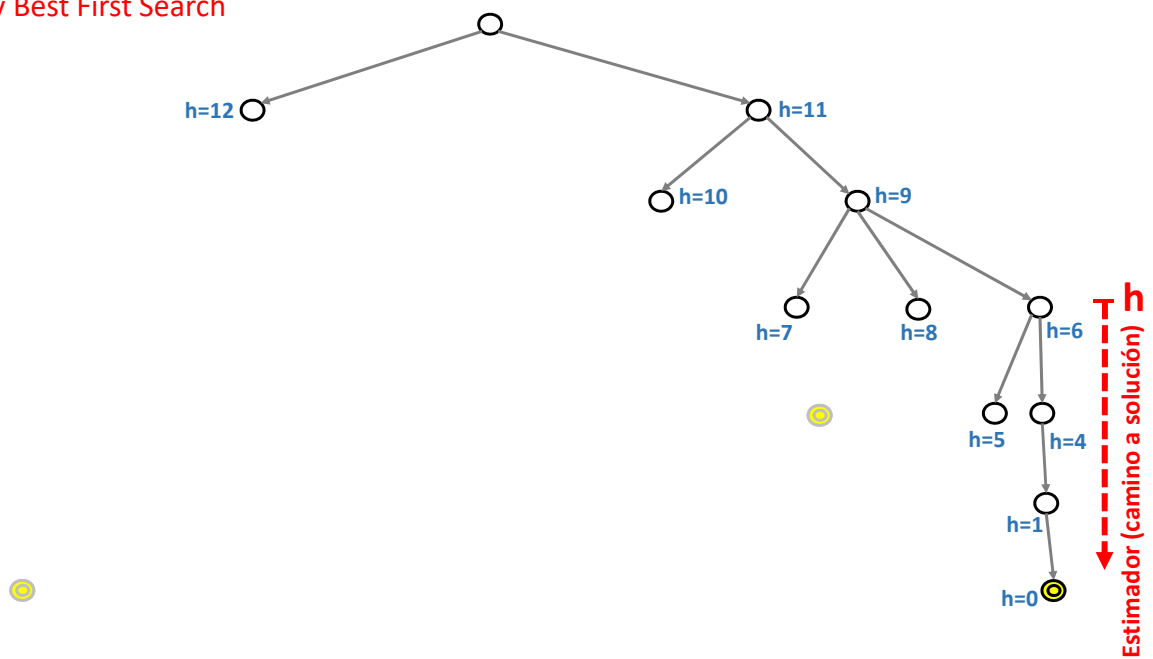
## Greedy Best First Search



## Greedy Best First Search



## Greedy Best First Search





## Greedy Best First Search

